Semiconductor 101

Learning Outcomes

Upon completion of this workshop, we should be able:

1. Compare the End-to-End Semiconductor Production Process

2. Compare Global Semiconductor Markets

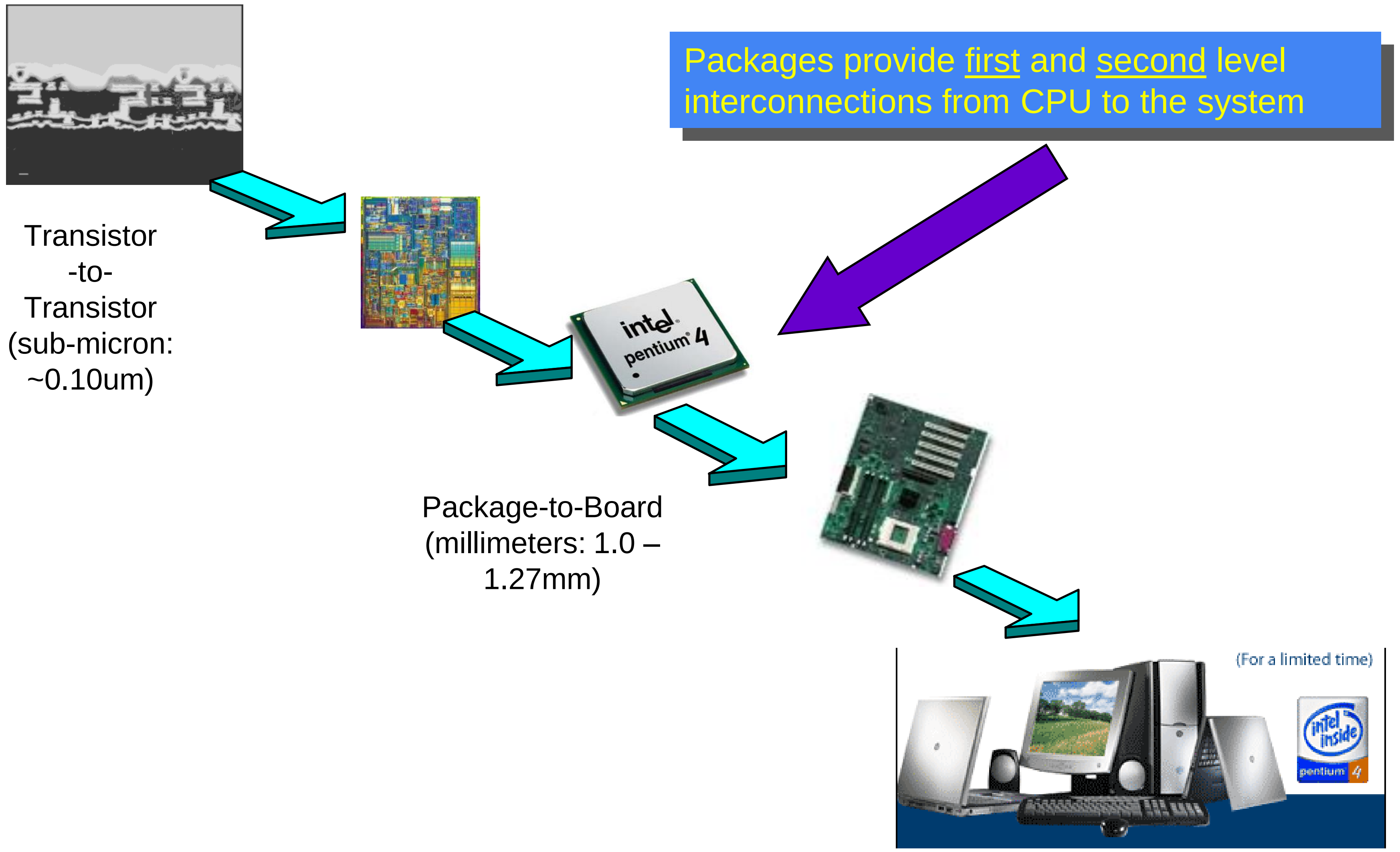
3. Analyze Semiconductor Industry

Introduction

- Introduction to the Semiconductor Industry
- Nature of Semiconductor Production
❖ Design, Fabrication, ATP, Equipment
- Semiconductor Value Chain
- Semiconductor in Comparison with the Global Semiconductor Industry
- Semiconductor Challenges

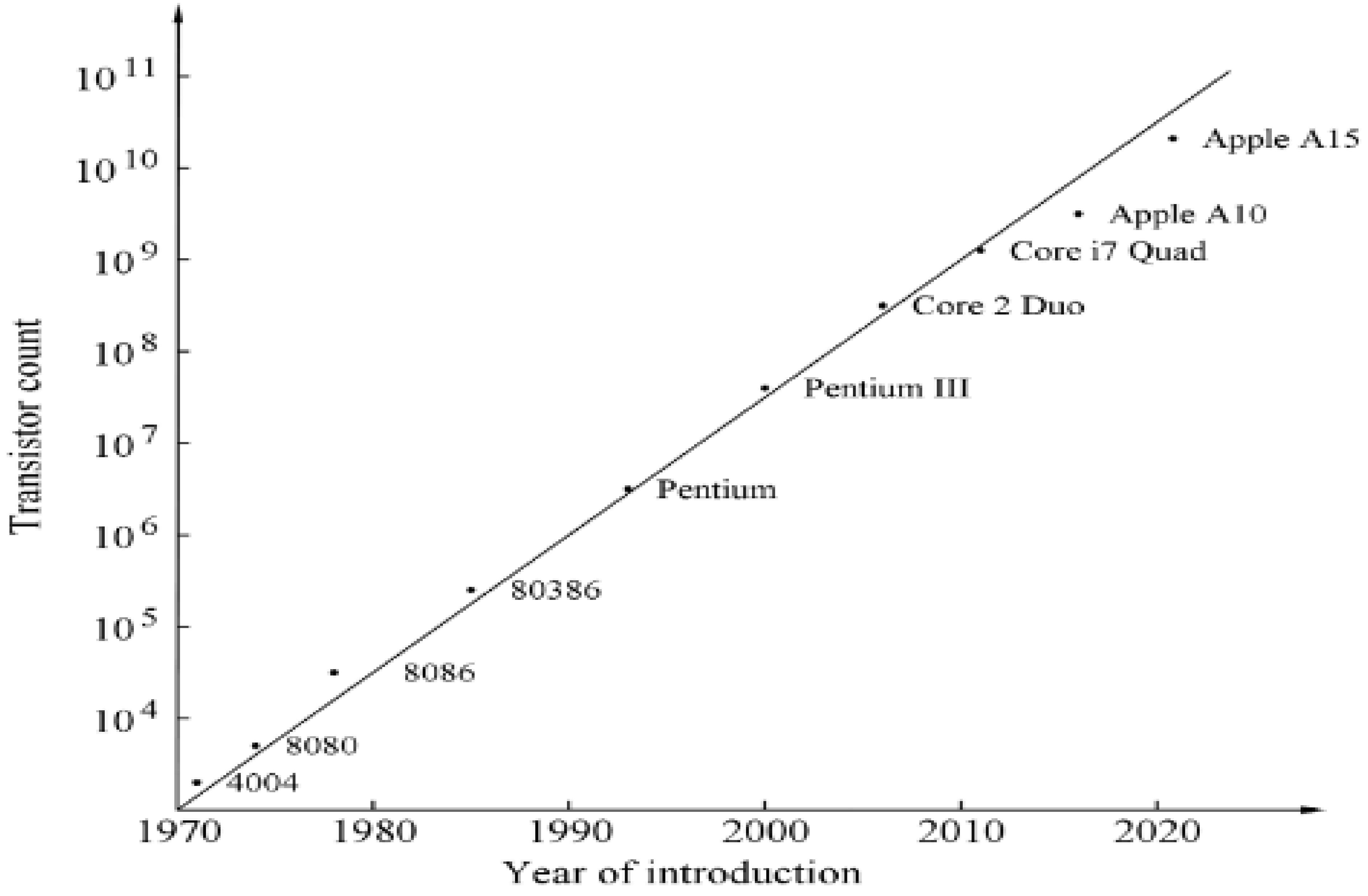
Introduction to the Semiconductor Industry

System & Interconnect Hierarchy



system (e.g. computer)
Functional Units (e.g. Register, memories, ALU, etc)
Gates and Flip Flop
Electronic circuits
Components (e.g. transistor, resistor, capacitor)

The semiconductor economic



Moore's Law: This principle, attributed to Intel co-founder Gordon Moore, states that the number of transistors on a microchip doubles approximately every two years, leading to an exponential increase in computing power.

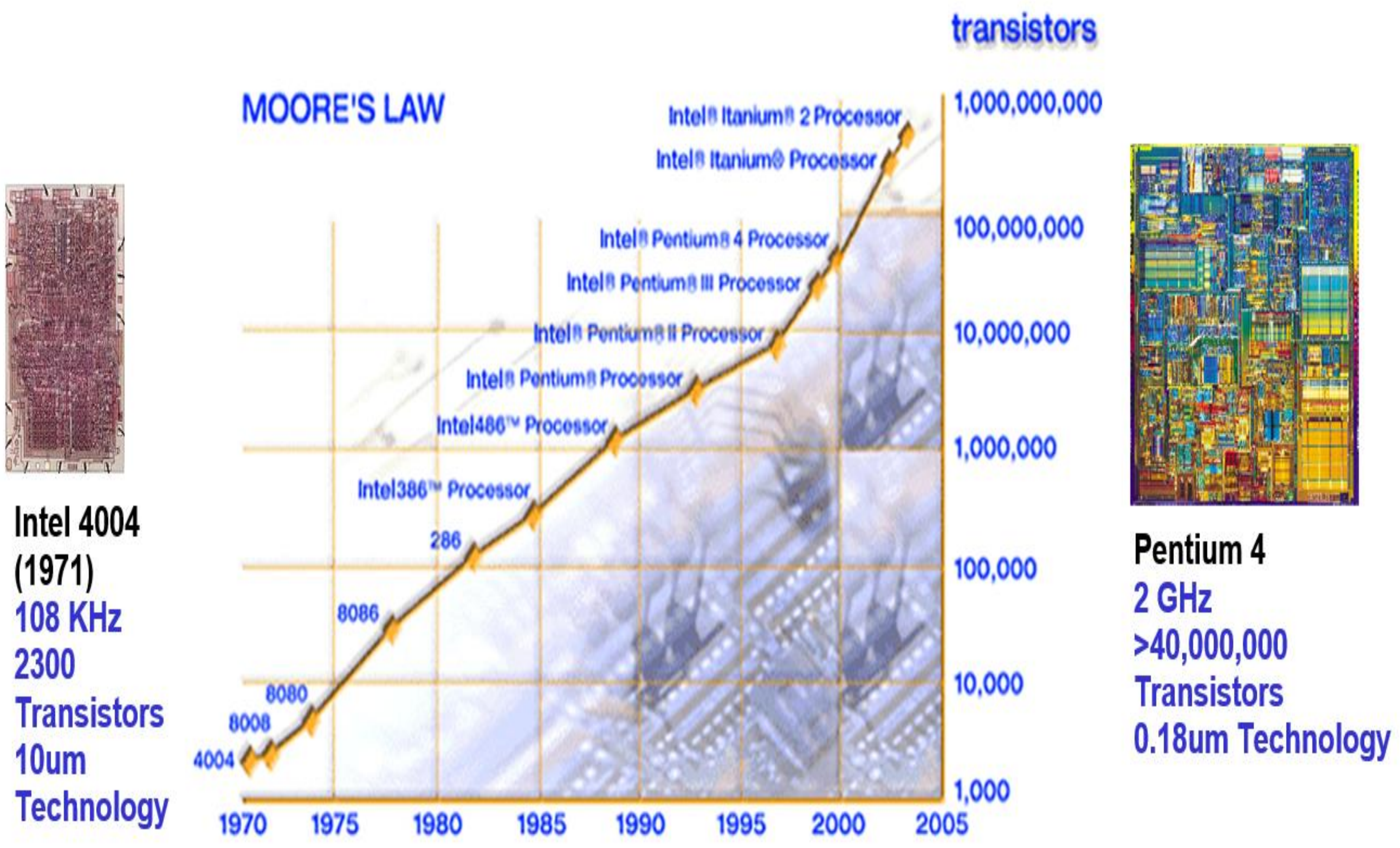
Another view

1 wafer = 1000 chips @ year 1 (technology 1)

1 wafer = 2000 chips @ year 3 (technology 2)

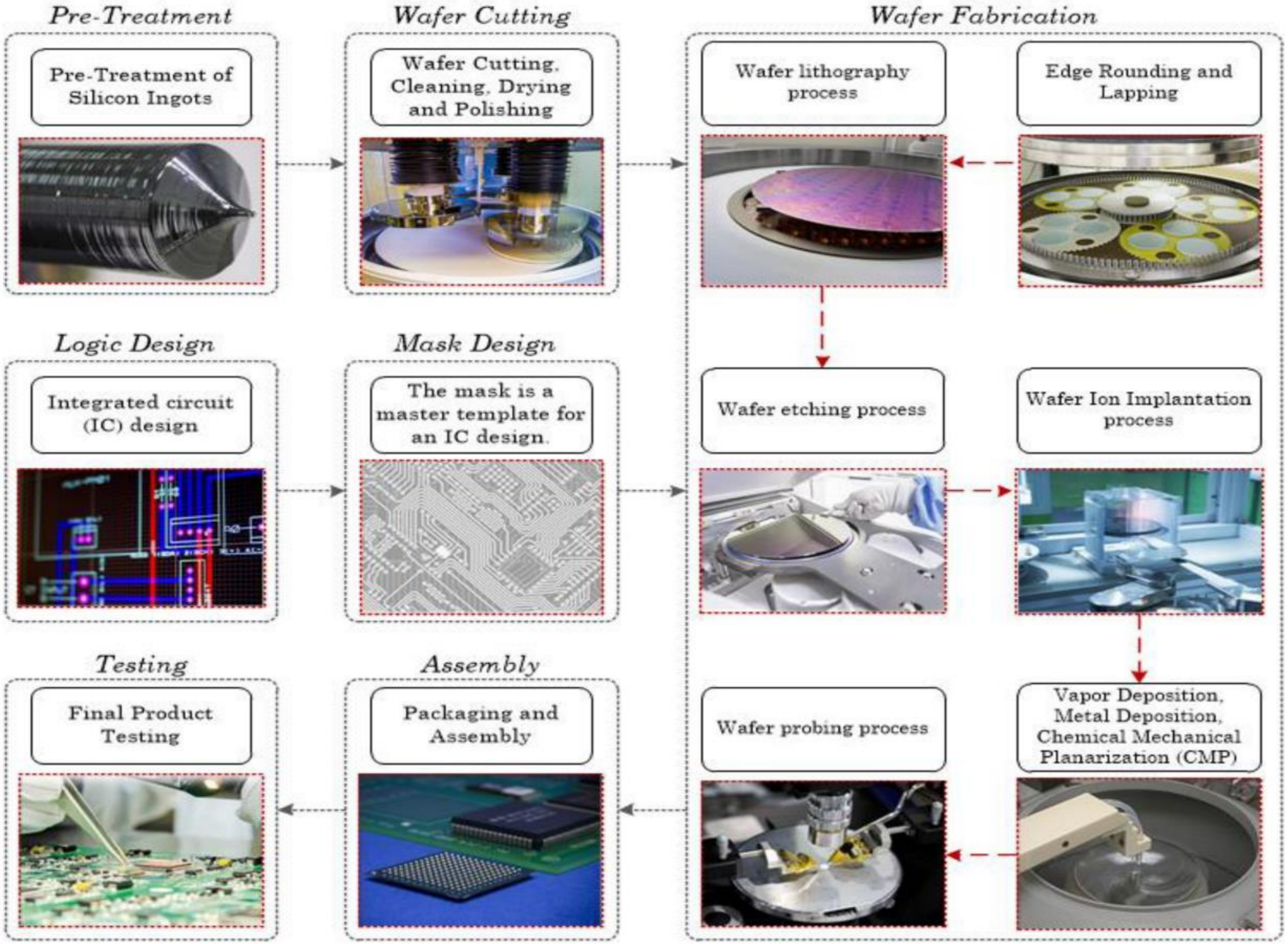
The key is that we manufacture/fabricate thousand of chips at One cost. A cost-sharing strategy. Definitely for mass consumption.

The semiconductor economic

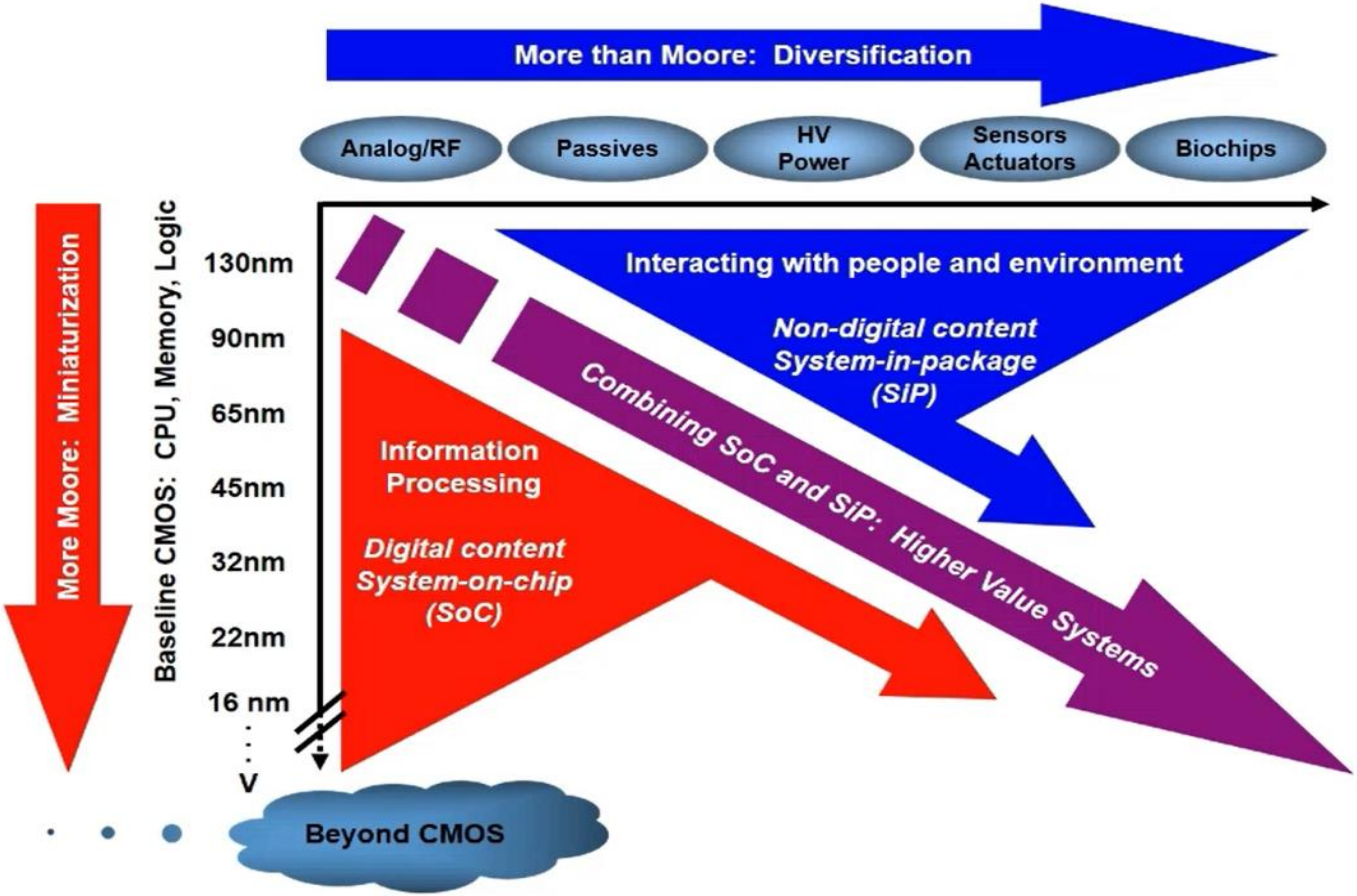


- Result of increasing components or circuit density per chip is the lower cost per component.
- The basic assumption, of course, is that the increase in the cost of chip fabrication is less than the increase in the number of components.
- This also means that the rate of increase of components (and the function) is greater than the rate of increase of the cost per chip.

Overview of the semiconductor chain

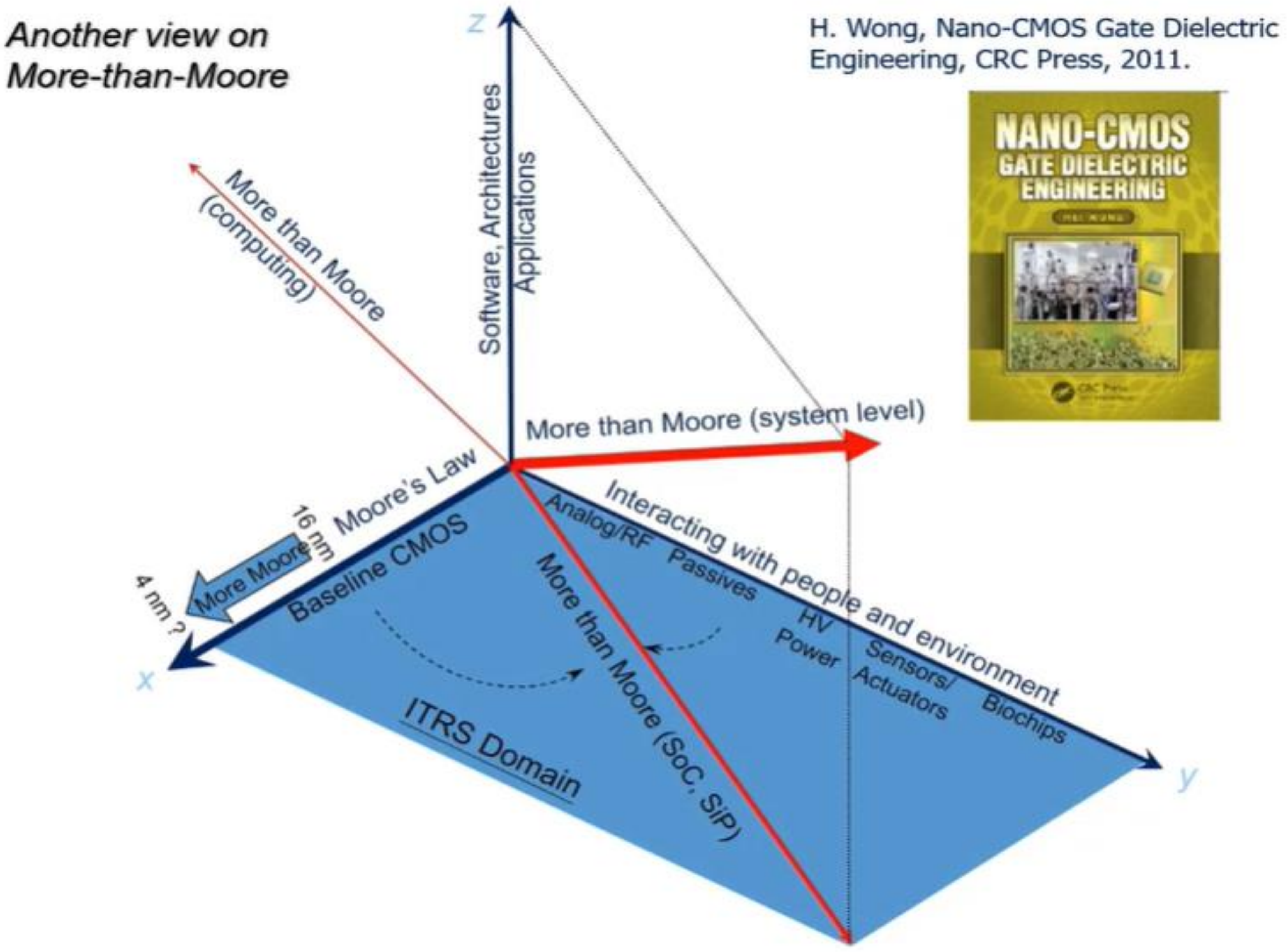


More than Moore



From SIA International Technology Roadmap for Semiconductors, 2010.

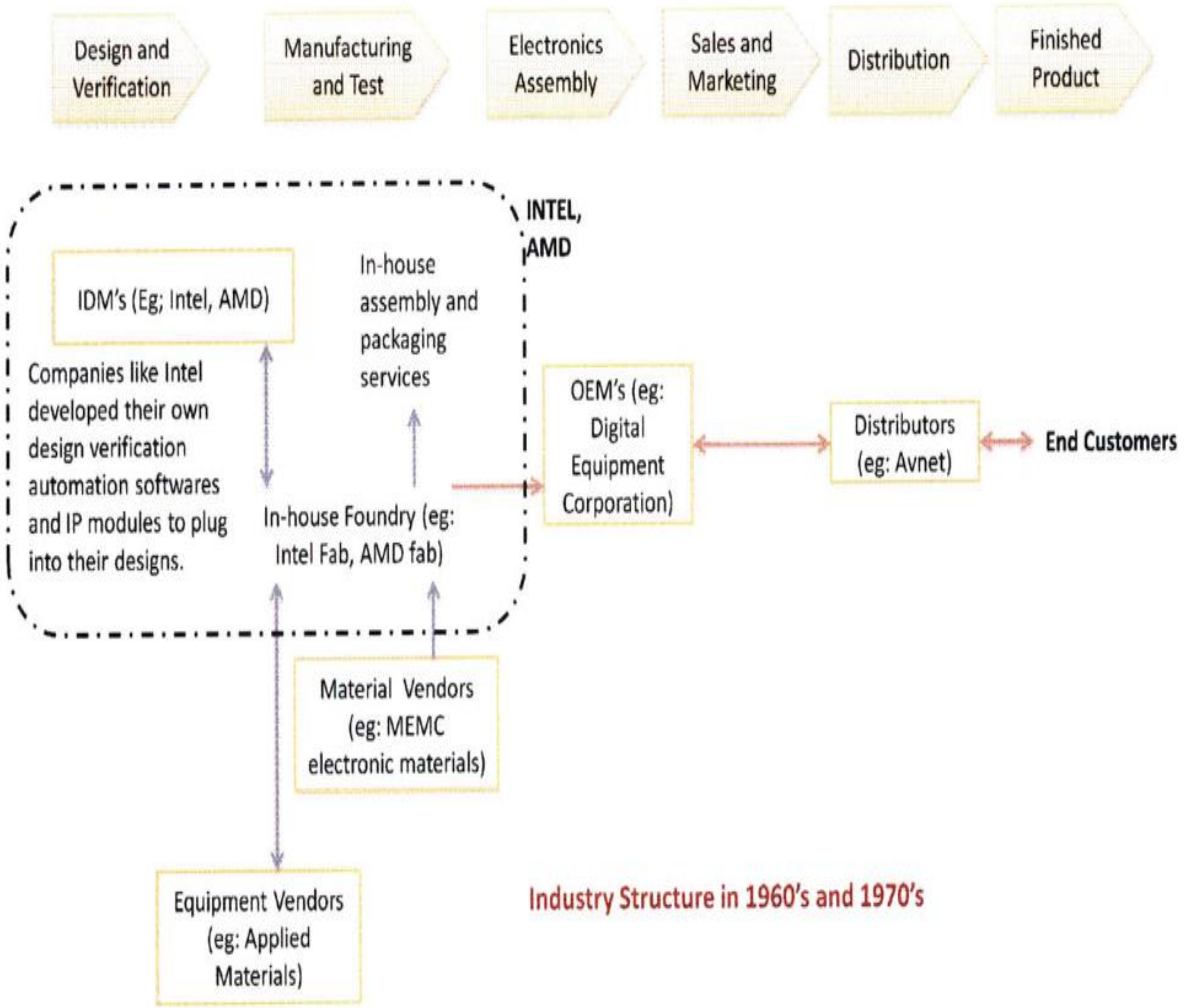
Another view on More than Moore



Nature of Semiconductor Production

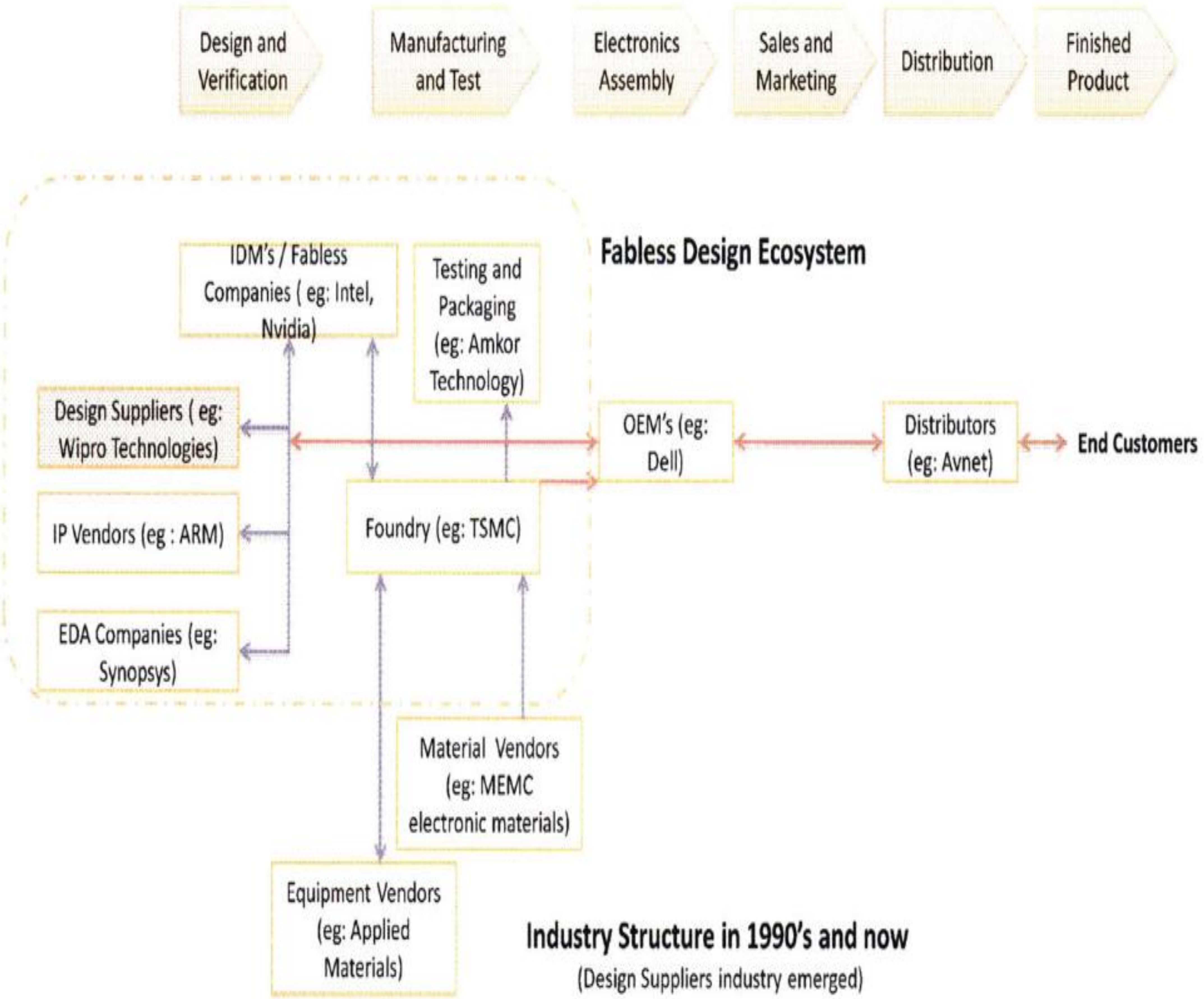
- Design, Fabrication, ATP, Equipment

Integrated Device Manufacturer



IDM is an integrated firm conducting chip design, production, marketing, supply chain management and after sale support.

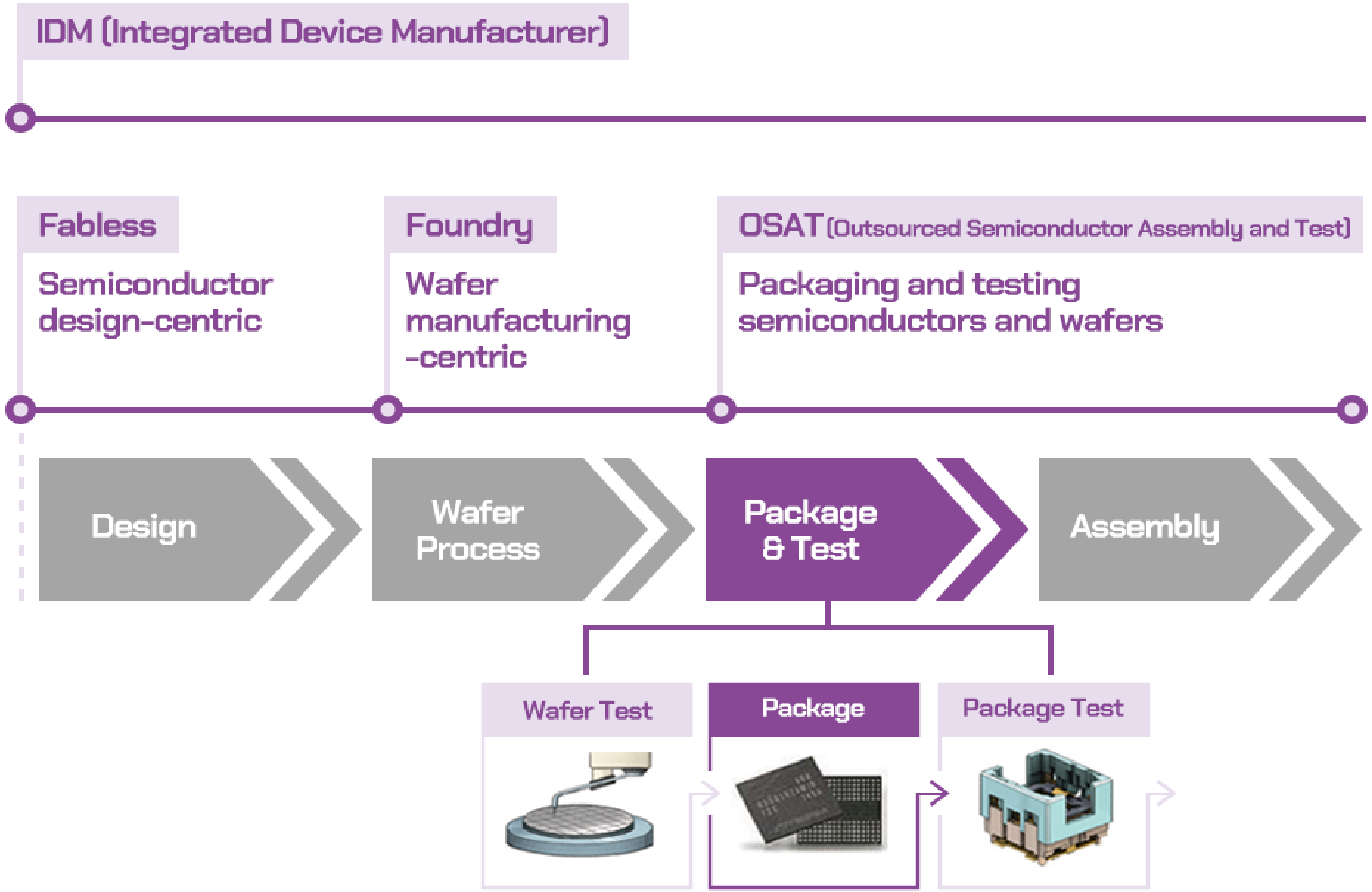
Fabless Company



The IC industry started a disruptive structural change from integration towards disintegration with the emergence of the foundry business in the mid-1980s

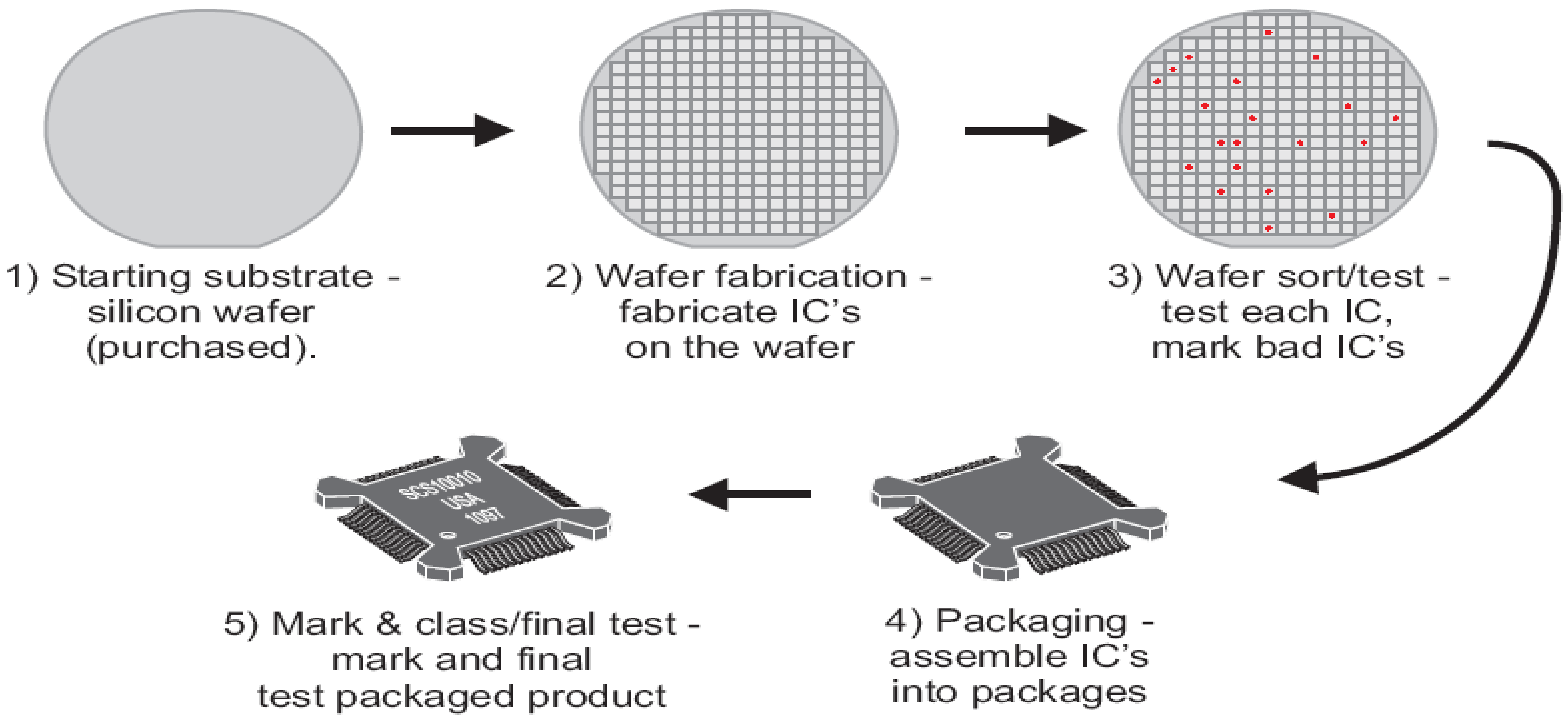
Ref:Outsourcing Trends In Semiconductor Industry
By
Karthikeyan Malli Mohan

The relationship between the semiconductor manufacturing process and the semiconductor industry (Source: Hanol Publishing)



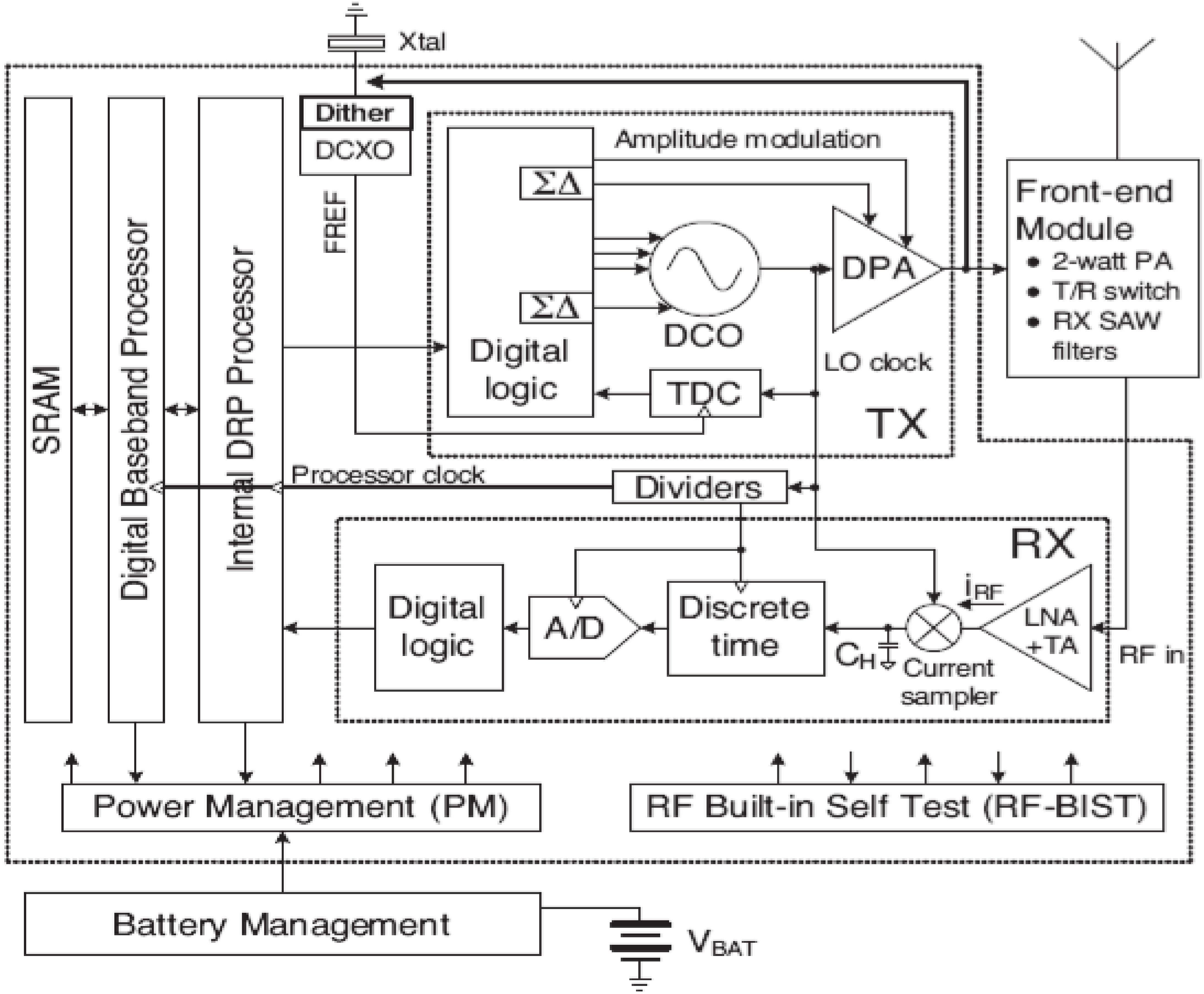
BIG companies like INTEL, Micron would have plants at different part of the world to reduce cost.

Integrated Circuit Manufacturing Overview



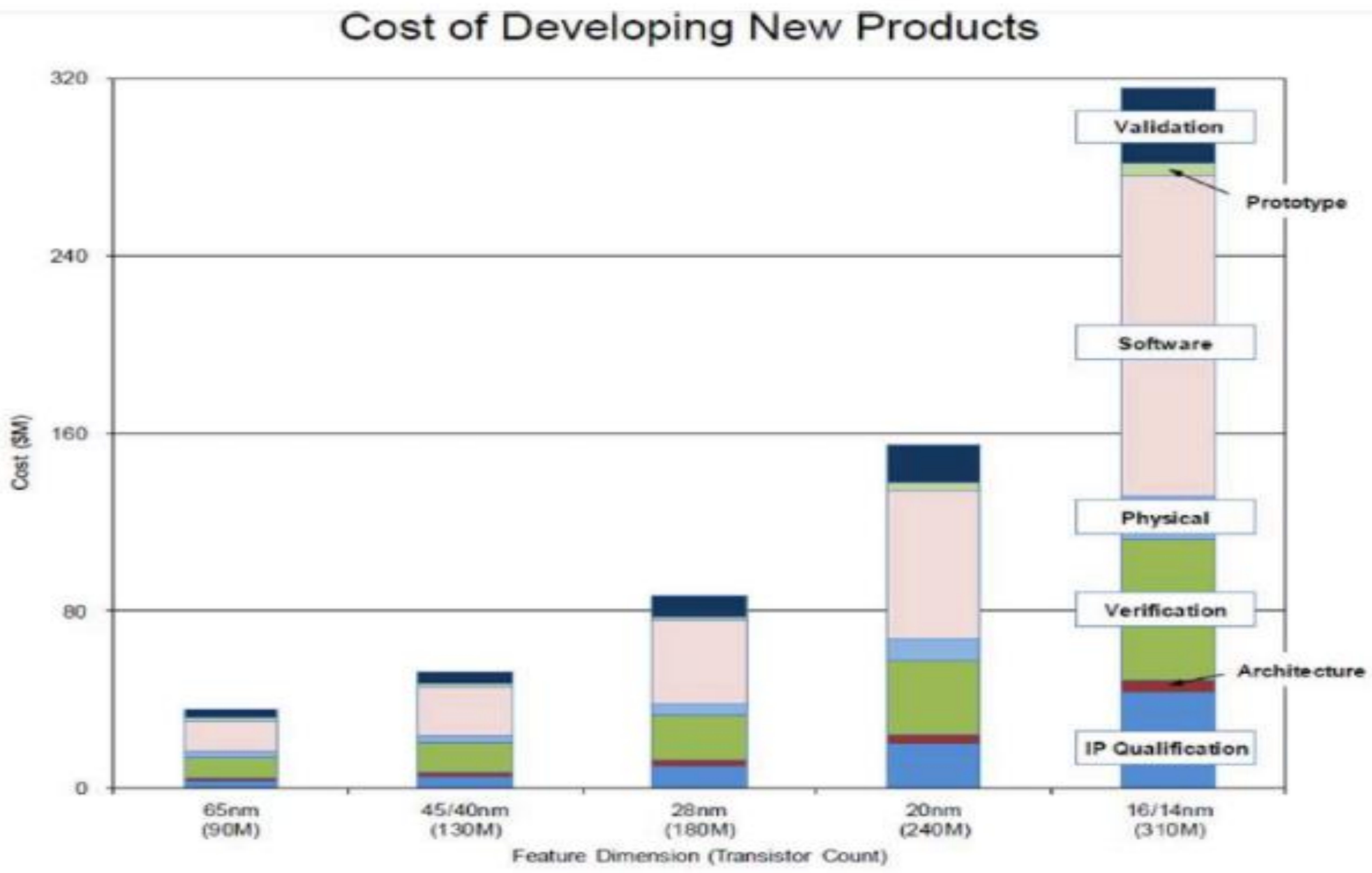
Design

Design Ideas in blocks diagram



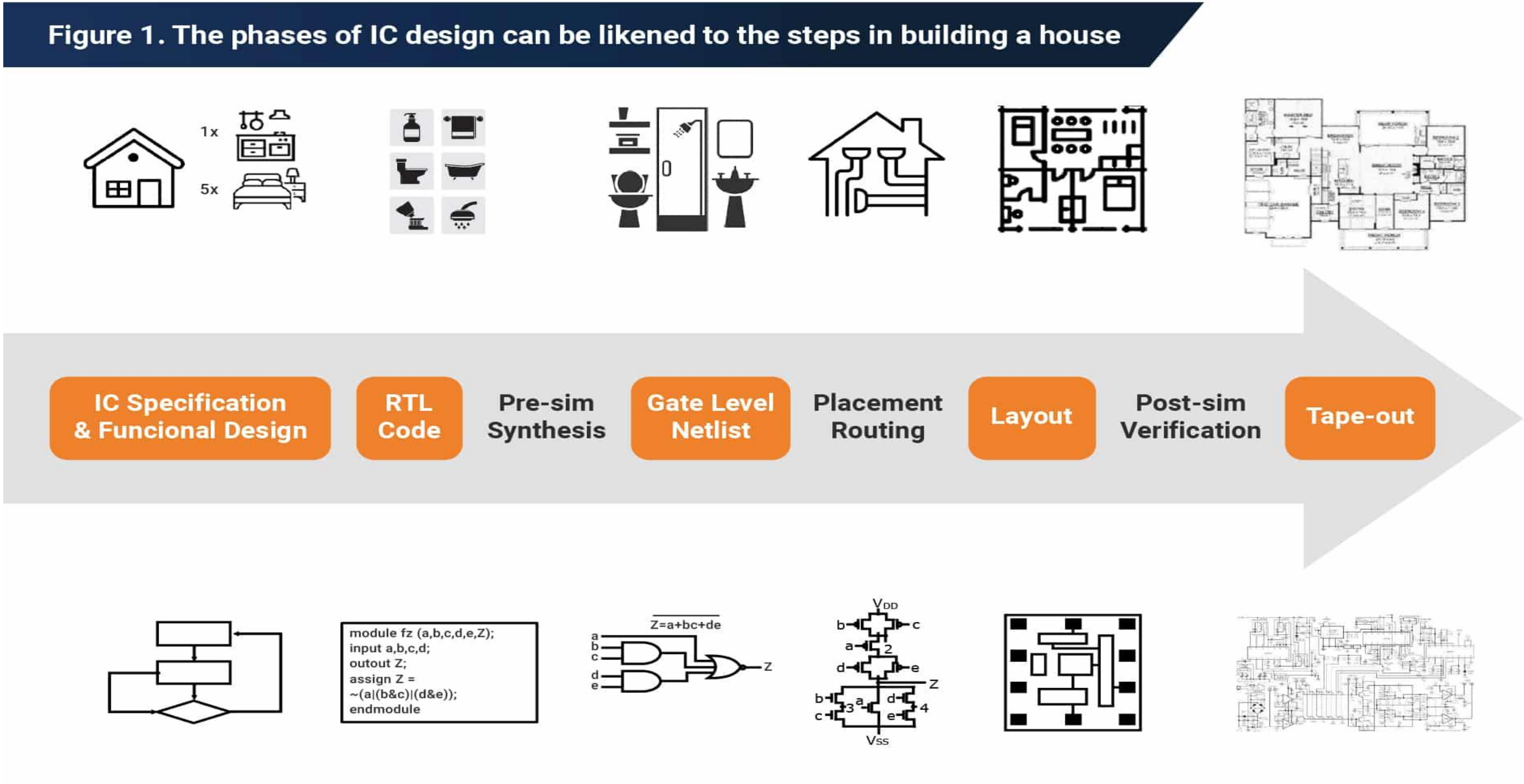
In the design process, companies conceive new products and specifications to meet customer needs and use these ideas to create particular logic and circuit designs for manufacture.

Cost

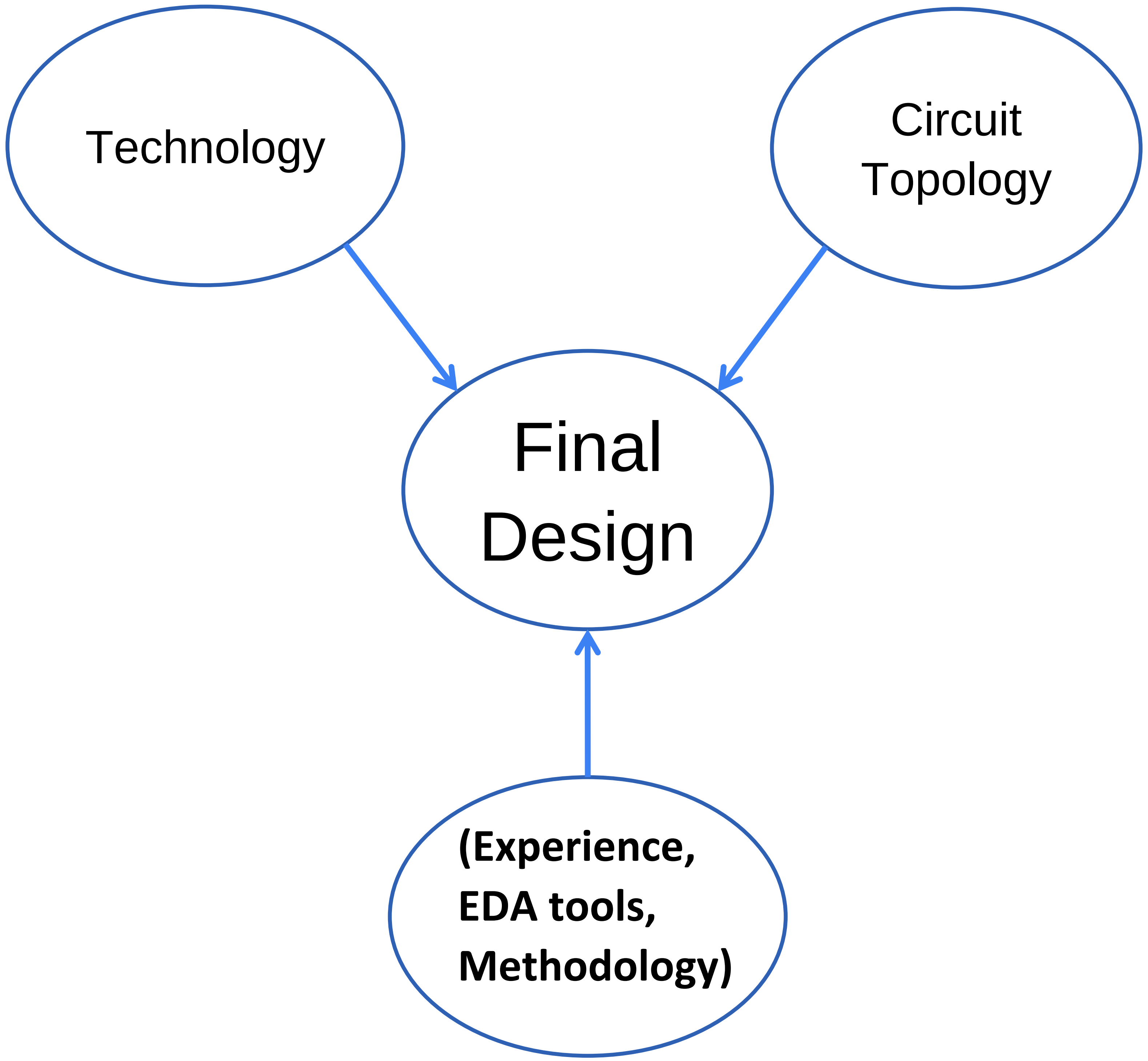


Cost of Developing New Products
320
240
160
80
0
Cost ($M)
Validation
Prototype
Software
Physical
Verification
Architecture
IP Qualification
65nm (90M)
45/40nm (130M)
28nm (180M)
20nm (240M)
16/14nm (310M)
Feature Dimension (Transistor Count)

Design Flow (FRONT END, BACK END)



Design Dimensions



- One can foresee the "novelty" of the final design by knowing the level of the "maturity" of these three factors.
- To "commercialize" an integrated circuit as a product, the level of maturity should be high.
- The lack of maturity can be considered as research gaps for the researchers.
- 'Novelty' could avoid IP infringement.

IP Vendors

- One of the quickest way to design a chip is to not to design it all The availability of design blocks from one technology to another can be a major asset in reducing time to market, cost and risk.

- Intellectual property can provide rich royalties for the patent owners in parts of the world where patent laws are strictly enforced. Qualcomm, inventor of CDMA using spread spectrum, command significant royalties within the mobile communications industry and ARM Inc is a major success story in capturing value through intellectual property in the microelectronic industry.

- It could be said that foundries take a leader role for the change in production ecosystem when international EDA and IP providers, such as Synopsys, Cadence and ARMs, play leading roles in proven IPs in the IP ecosystem.

Chiplet

In this process, functions previously performed by a single chip are divided into discrete functions and fabricated into smaller building blocks, so that only those blocks that need the highest performance (e.g., the logic component) use the most advanced process nodes, while other blocks (e.g., input/output power) use mature process nodes. The chiplet fabrication process allows unique chiplets to be purchased from different vendors and packaged together using a “plug-and-play” approach to provide devices with enhanced functionality (that is, as long as they are designed to communicate with one another).

Chiplet Interface Circuit Standard

IEEE SA STANDARDS ASSOCIATION

P3468 - Chiplet Interface Circuit Working Group (CICWG)

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https://sagroups.ieee.org/3468/

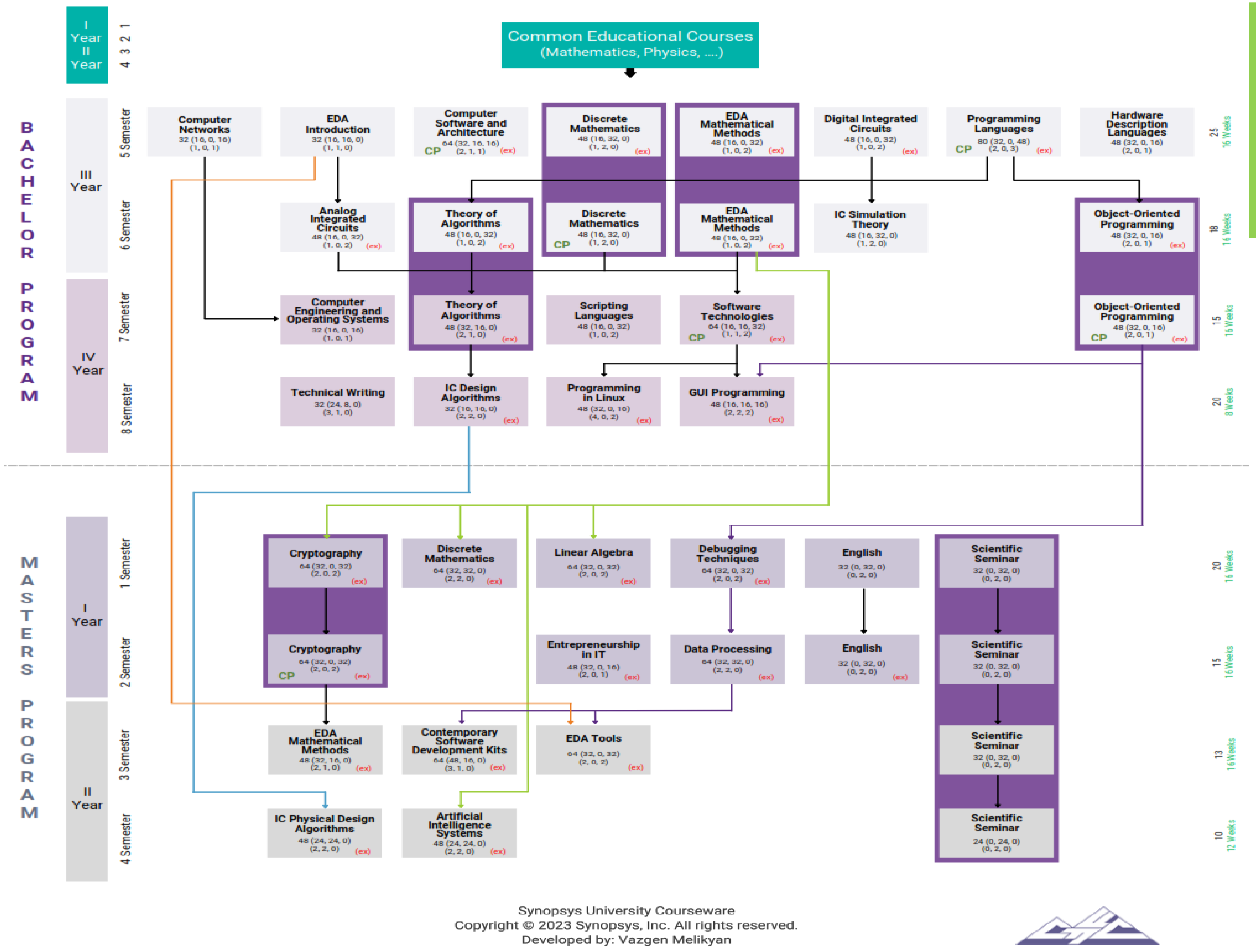
Expected Submission: Dec 2026

An industry effort is under way to standardize chiplet design to improve the ability to mix-and-match in the market and to enable an "open chiplet ecosystem" for end application developers to optimize hardware to their specific requirements.

EDA Tool

- EDA- ELECTRONIC DESIGN AUTOMATION; tools for the design– developed by SOFTWARE company.
- The company develop software to support various stages of IC design, including design entry, synthesis, simulation, verification, and physical implementation.
- Before the advent of these tools, integrated circuits were designed by hand and were manually laid out. These companies help semiconductor companies address the key challenges the designers and manufacturers are facing today and also help give a competitive edge in bringing their products to market quickly.
- The design service companies and the IDM manufacturers license software from EDA companies.

Synopsys EDA Course Plan



https://www.synopsys.com/content/dam/synopsys/community/sara-documents/eda-course-plan.pdf

Fabrication

Fabrication-Foundry of Fab

After the design stage, semiconductor chips are manufactured, or fabricated, in facilities often referred to as fabs or foundries. In this front-end fabrication process, chips are manufactured on circular sheets of silicon or, less commonly, other semiconducting materials, called wafers, that are typically about 8 or 12 inches in diameter. Each circular silicon wafer typically contains hundreds of different chips (usually tiny rectangles, smaller than the size of a postage stamp).

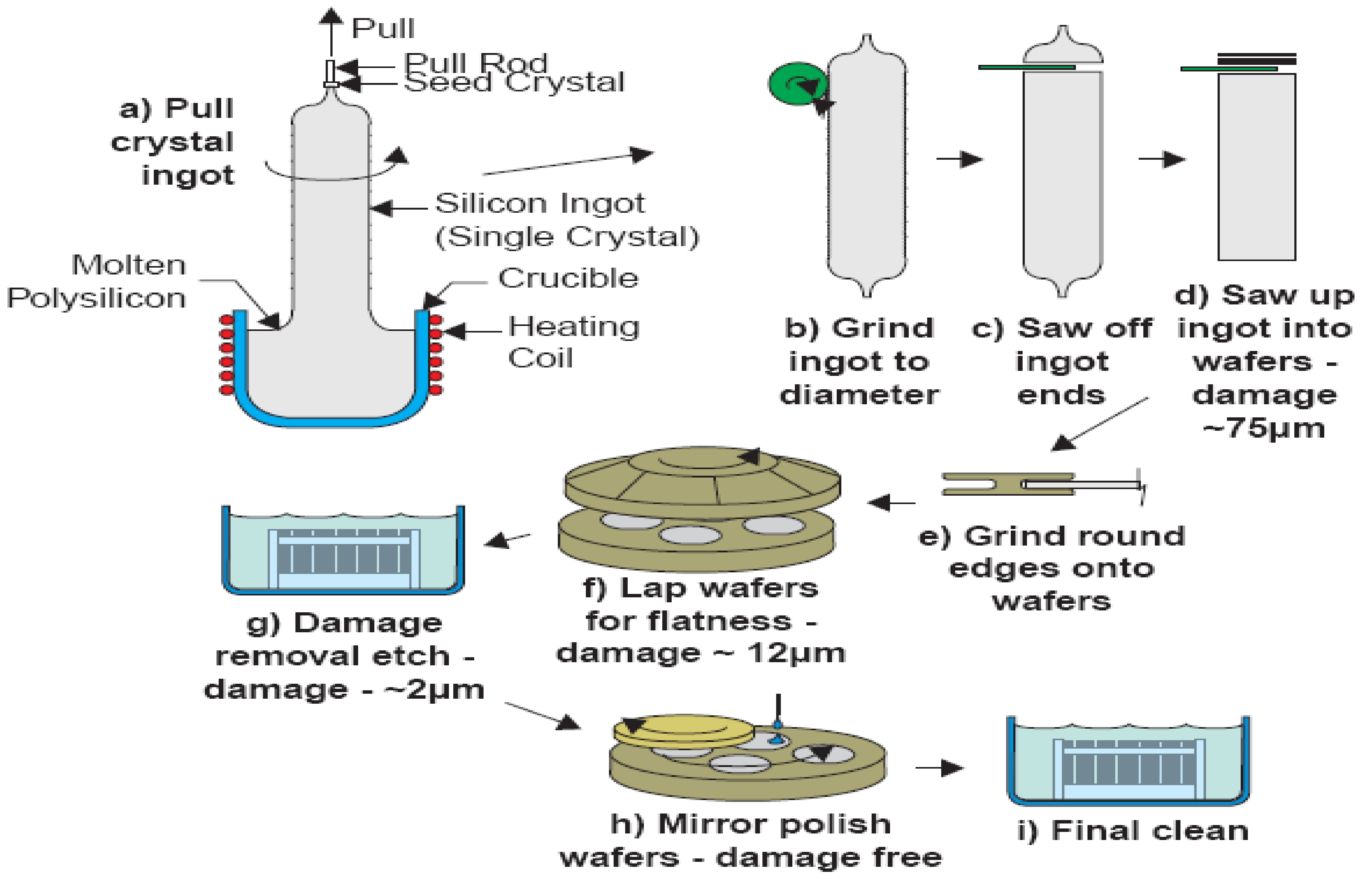
PDK: PROCESS DESIGN KIT; contains the technology information– developed by Foundry.

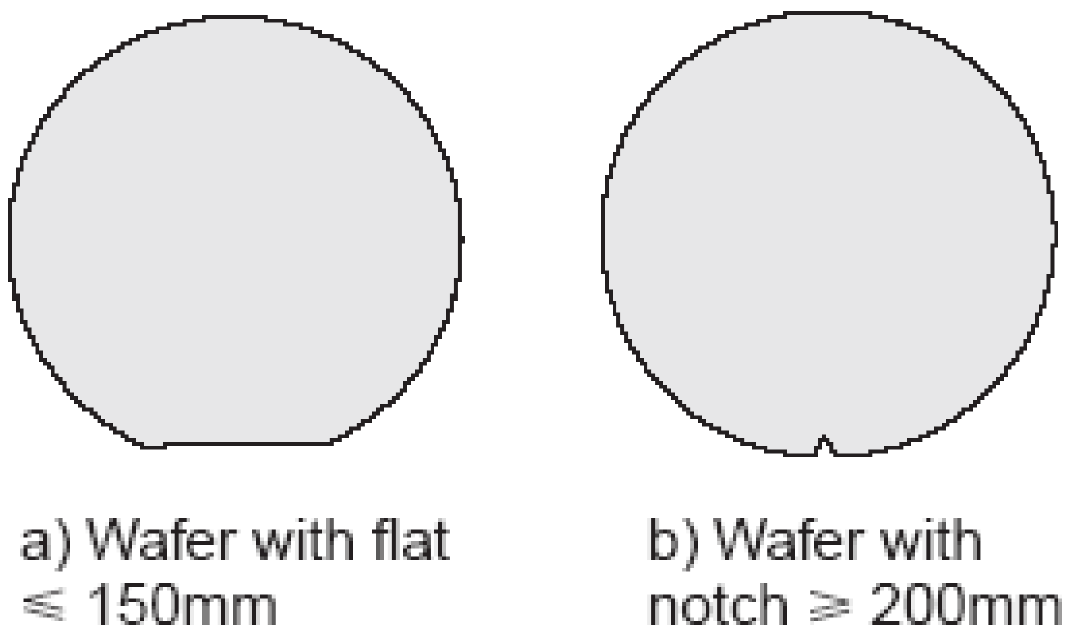
Restrictive and expensive PDK licensing by foundries can inhibit small and medium-sized innovators from realizing the production of their designs. Some companies are partnering to offer open-source PDK to new designers, to promote innovation and enable them to produce chips more quickly

Fabrication Terminology

- Fabrication involves creating ICs on a wafer made of semiconductor material, usually silicon.
- Photolithography: Patterns from the design are transferred onto the wafer using ultraviolet light.
- Doping and etching: Impurities are added to the semiconductor to change its electrical properties, and unwanted material is removed to create the necessary structures.
- Deposition and polishing: Layers of materials like metals or insulators are added and polished for connectivity between transistors.
- Wafer dicing: Wafers are cut into individual ICs (chips).

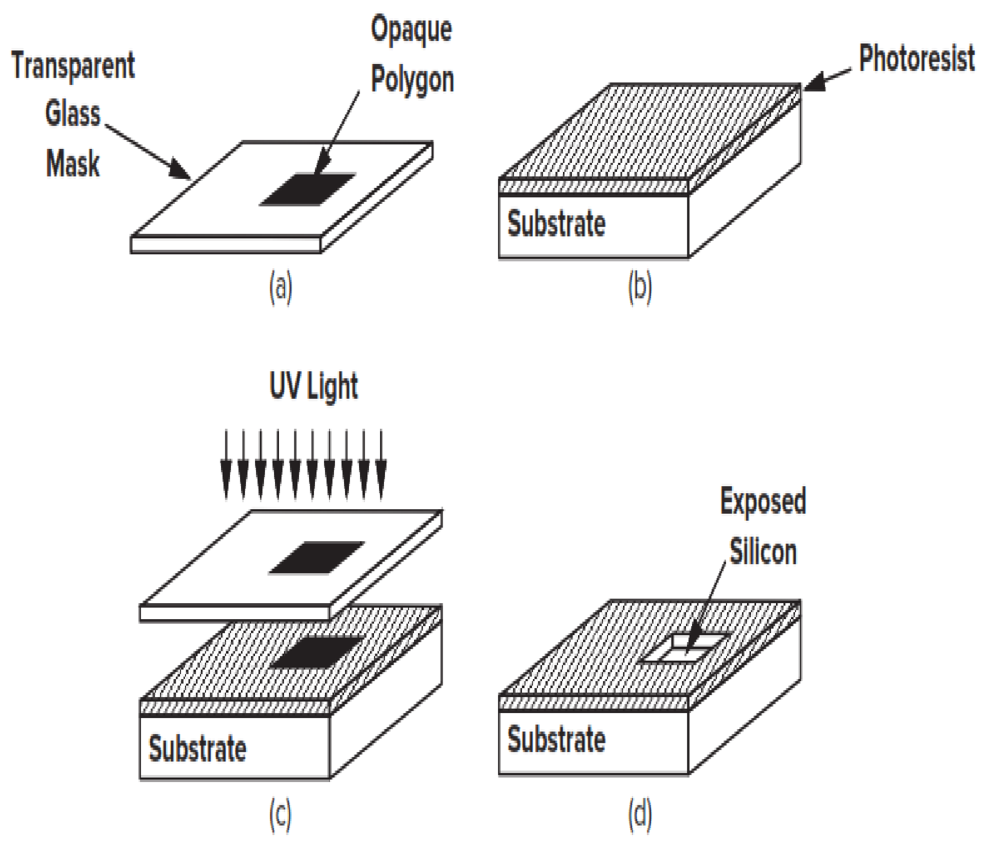
Silicon

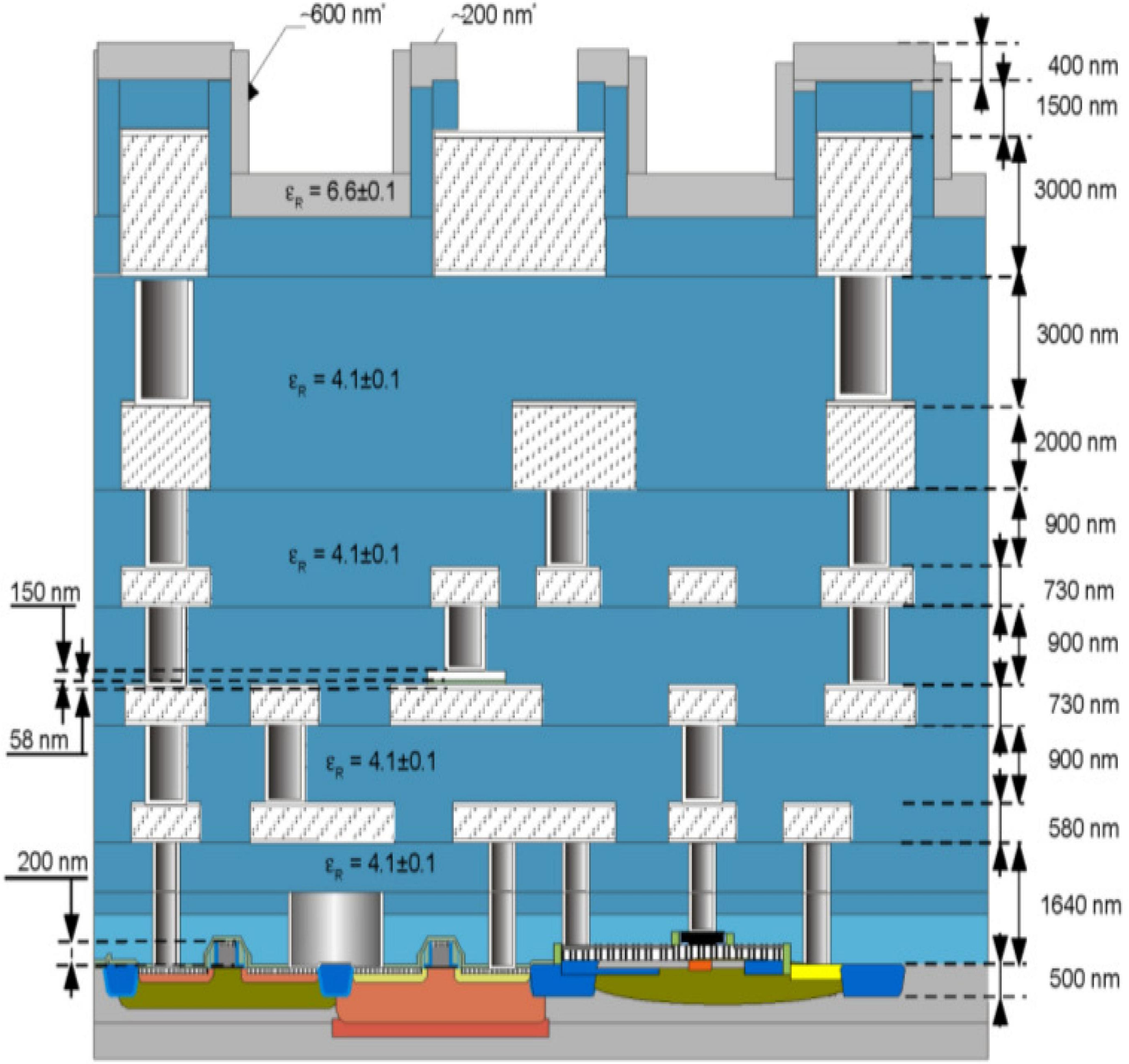




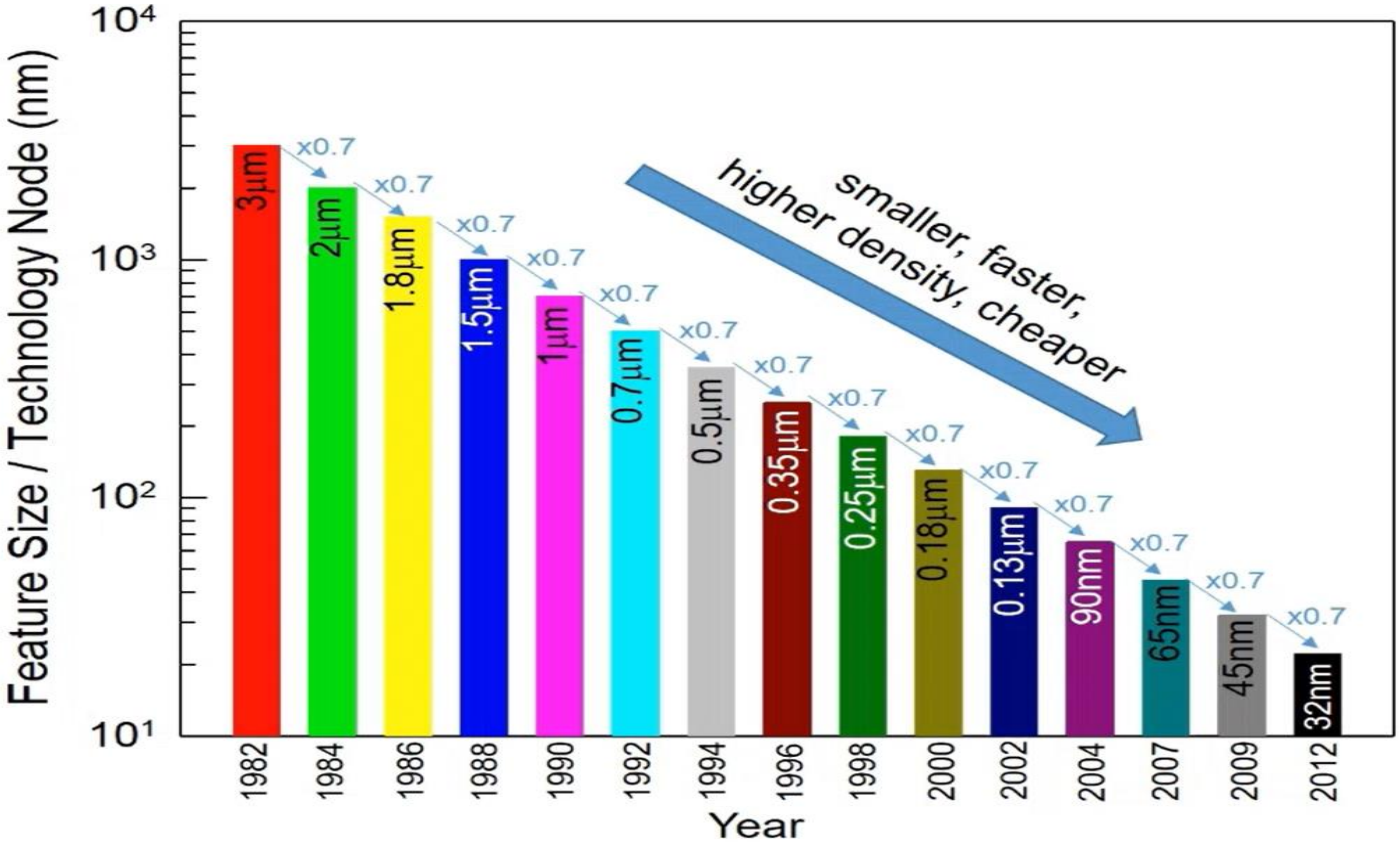
It normally has a few millimeters thick and anywhere from 100mm (4″) to 300mm (12″) in diameter.

Fabrication Steps

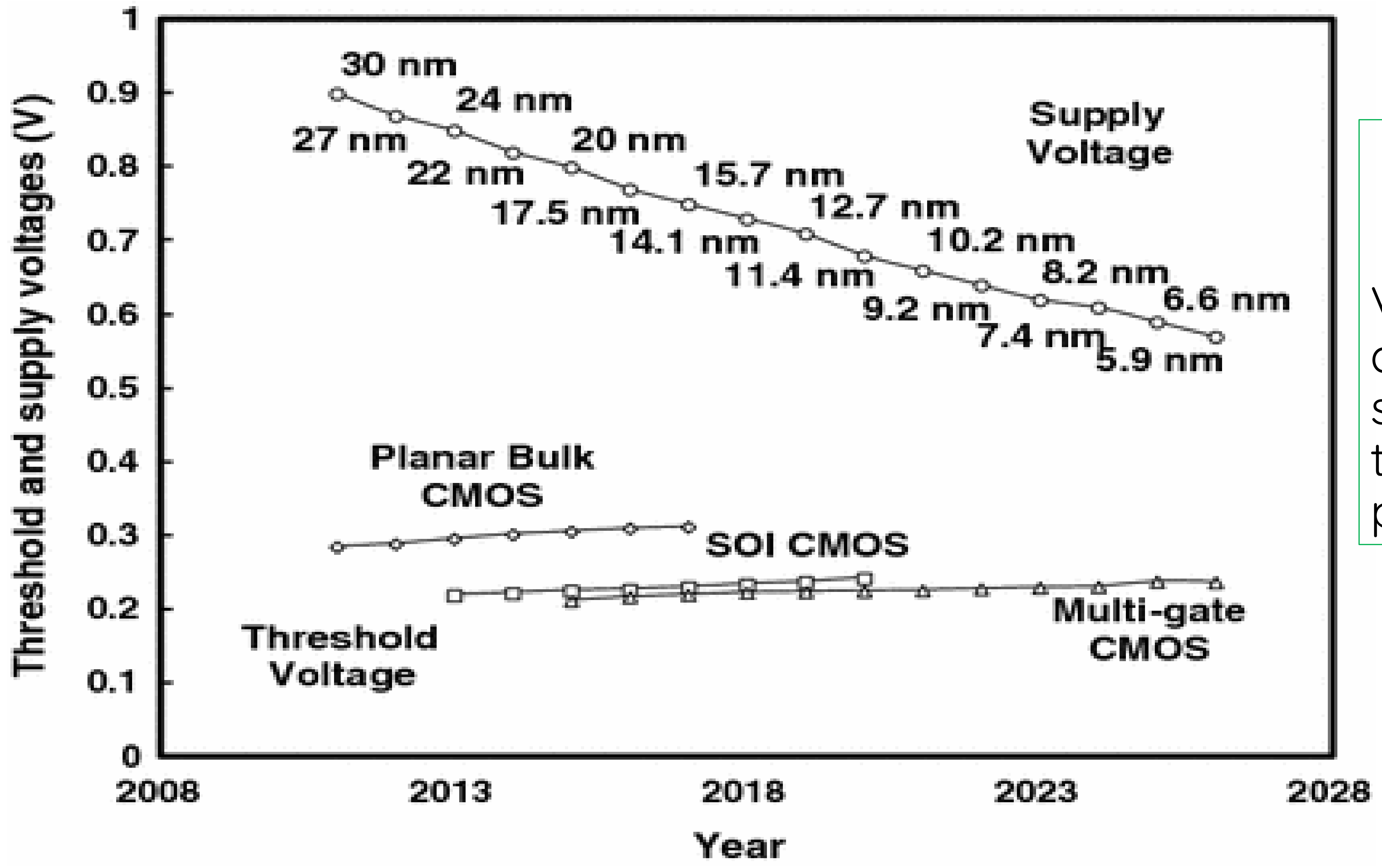




Scaling (ITRS)

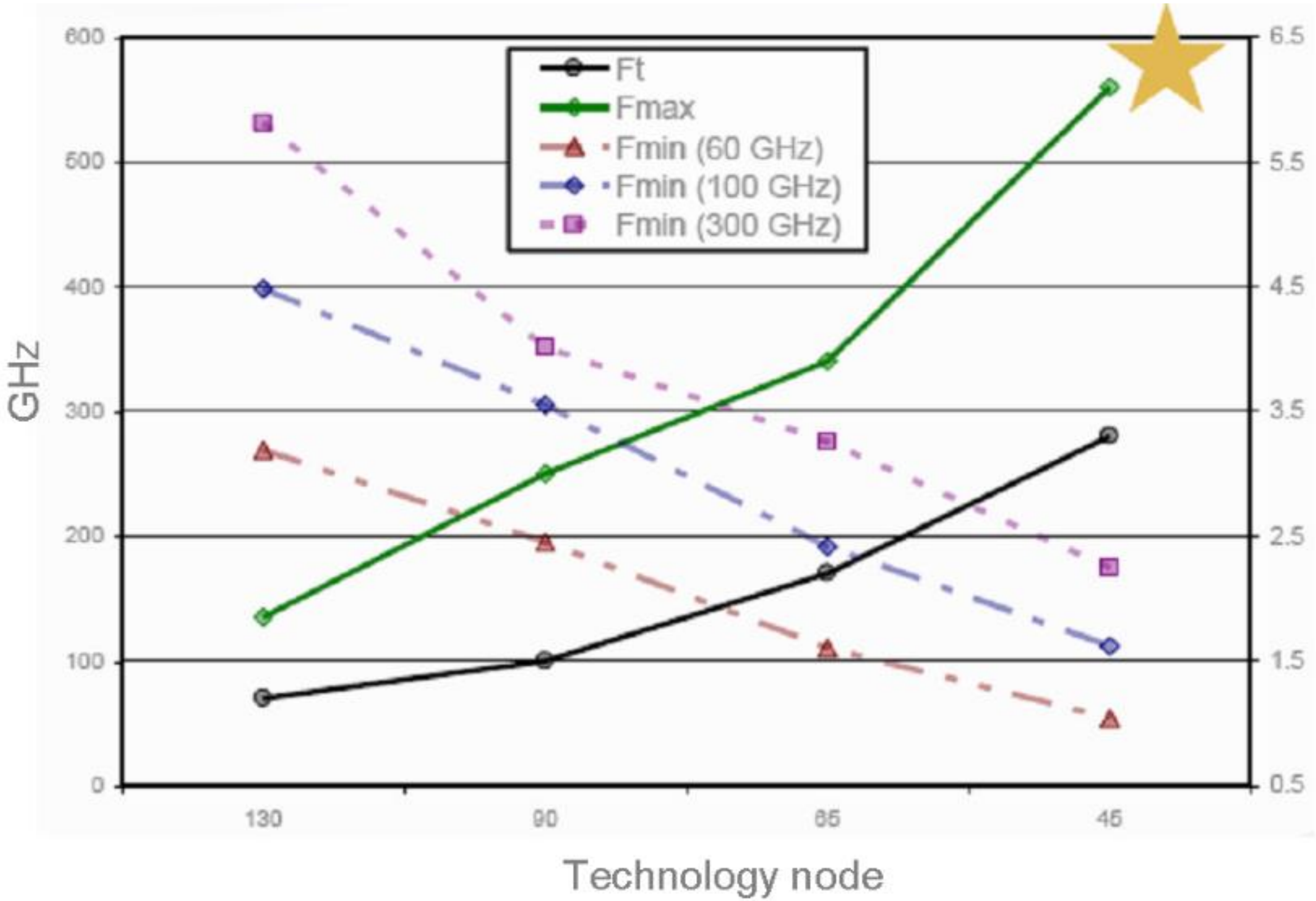


Scaling - Reality

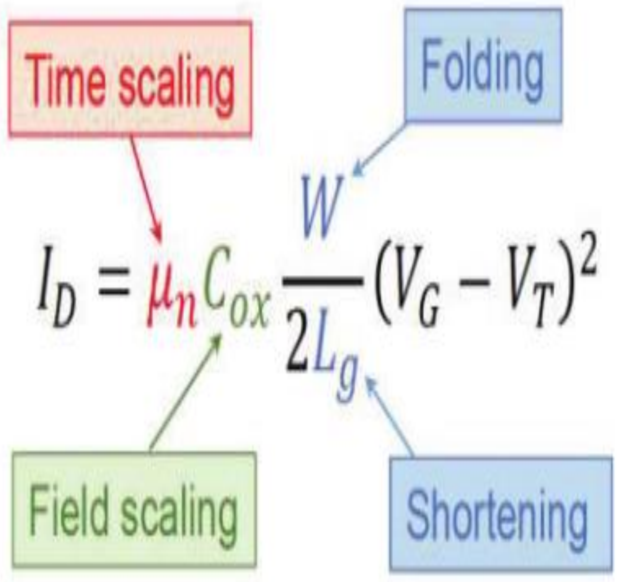


Vth is approximately the same because of the leakage problem.

Ft and Fmax vs. Technology node



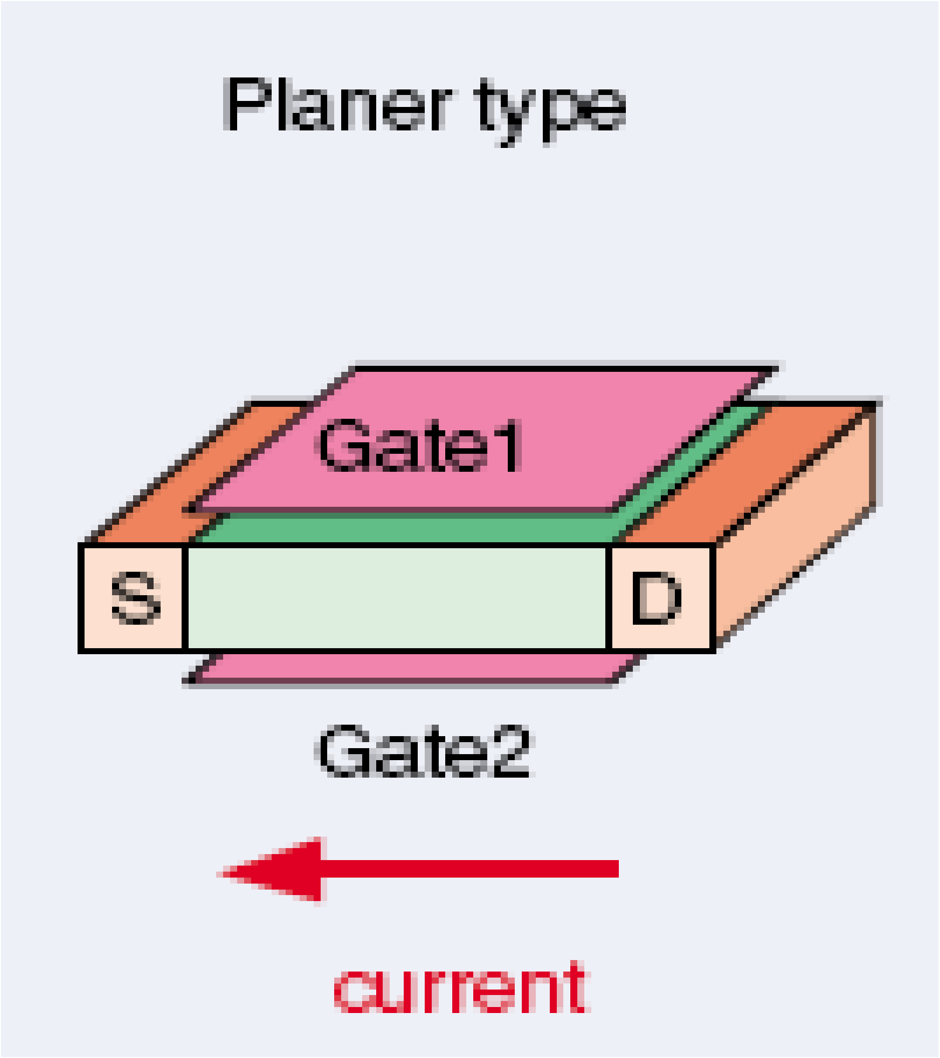
More Moore (Current solution)

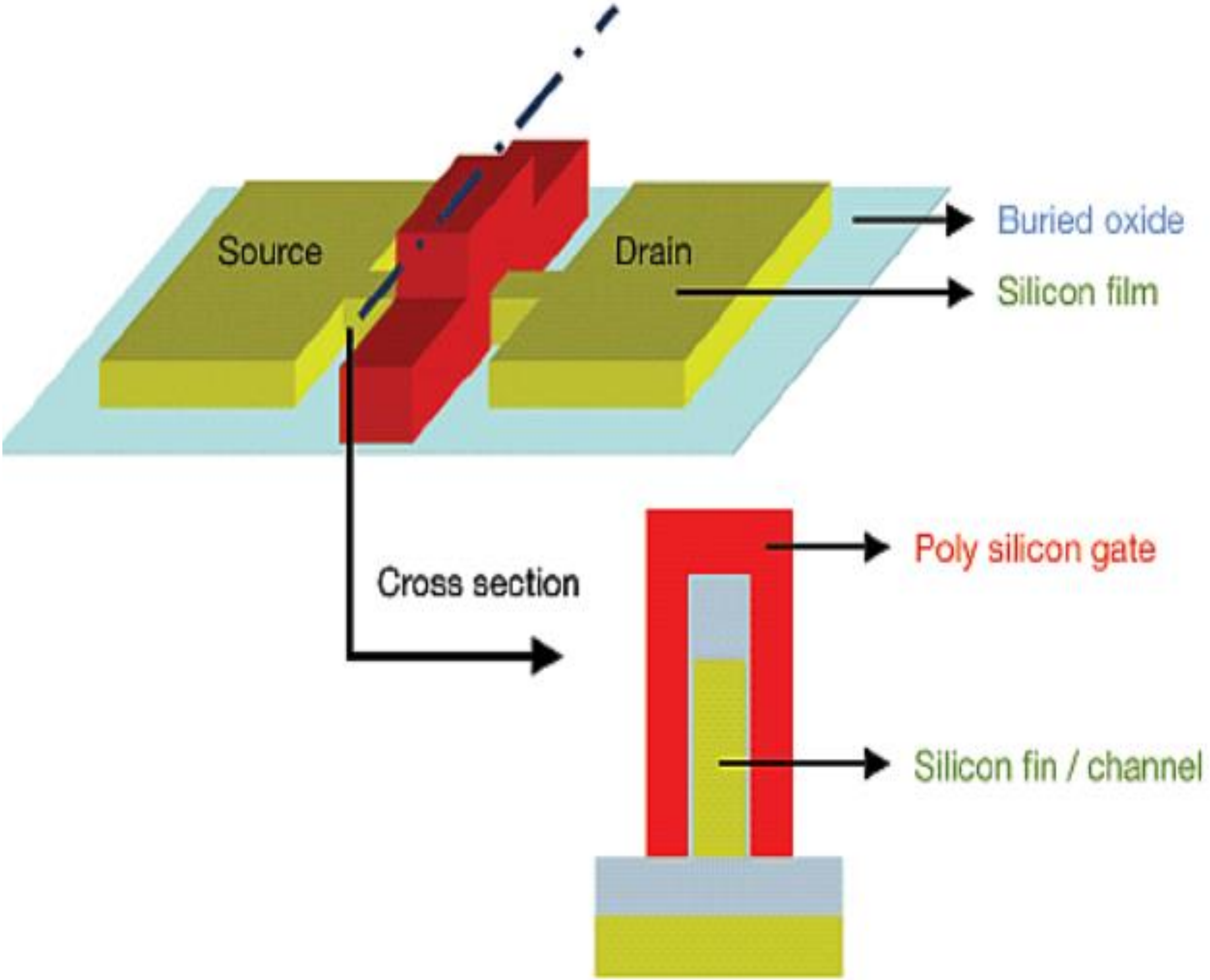


Technology Node	20/22 nm	14/16 nm	10 nm	7 nm	5 nm
Gate Length (nm)	25	22	19	16	10
Gate reduction factor	1	×0.88	×0.86	×0.88	×0.625
Fin Width (nm)	10	9	8	6	6
Fin Height (nm)	34	42	50	52	54
Fold Ratio*	0.128	0.097	0.074	0.055	0.054
Fold ratio reduction	1	×0.76	×0.76	×0.74	×0.98
Estimated Area Reduction	1	x0.67	x0.65	x0.65	x0.61
Density (MTr/mm²)	22.4 Apple A8	26.4 Apple A10 Fusion	44.6 Apple A10X	86.3 Apple A13	134.4 Apple M1
Density increment	1	118%	169%	193%	156%

$$*Fold\ Ratio = \frac{W_{fin}}{W_{eff}} = \frac{W_{fin}}{W_{fin} + 2H_{fin}}$$

Transistor Structure



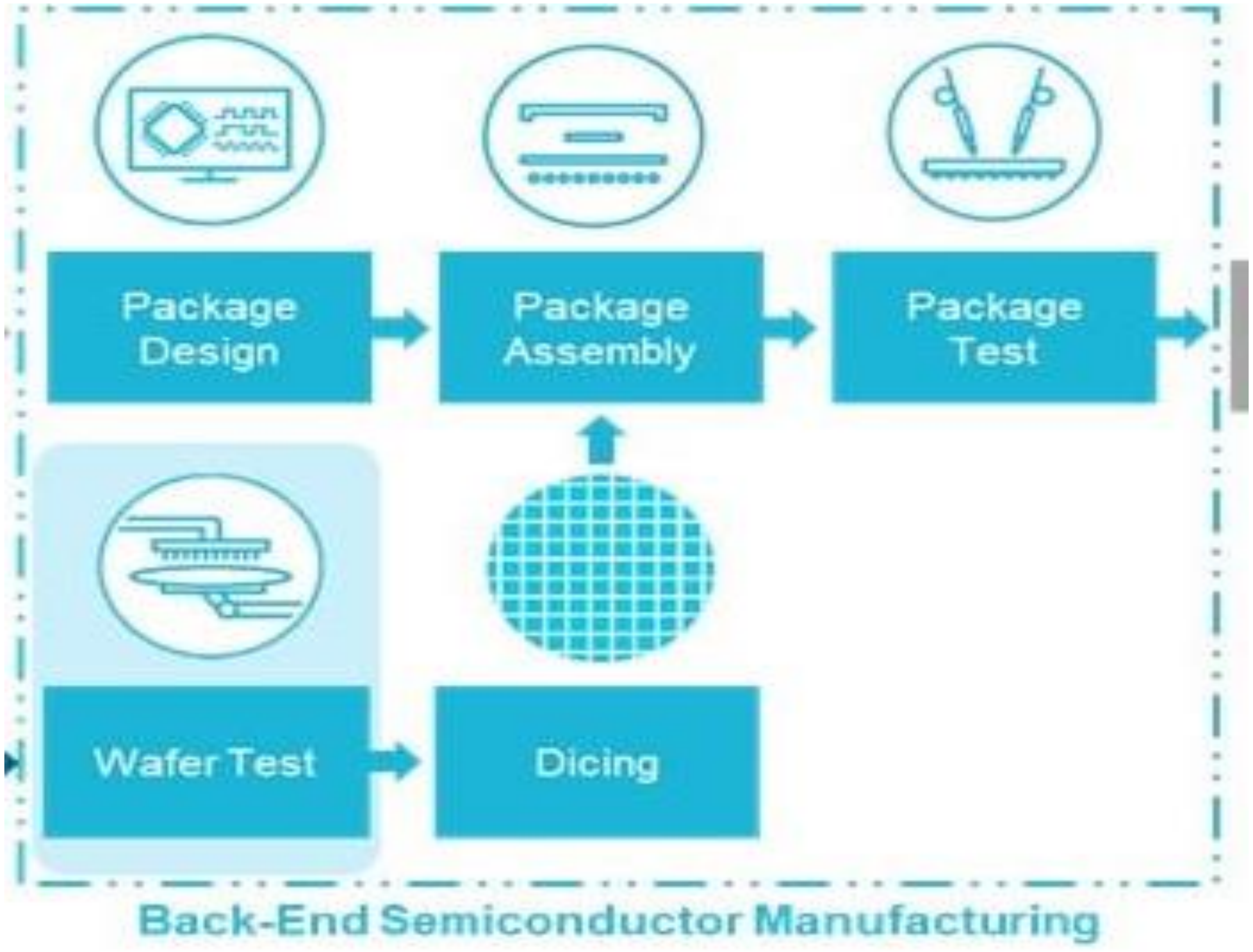


What Wafer Foundry need

Magnitude of Resources:
•**Electricity Consumption:** A single large semiconductor fab can consume electricity equivalent to tens of thousands of homes. For example, a modern fab can use over **100 megawatts (MW)** of power, enough to power a small city.
•**Water Consumption:** A large fab can use **2 to 4 million gallons of water per day**. This is comparable to the water usage of a large town. Water is continuously cycled through the system, but still, large quantities are consumed, especially in areas without efficient recycling systems.
Environmental and Economic Considerations:
•**Sustainability Efforts:** The environmental impact of such large energy and water use has driven many companies to invest in renewable energy sources, energy efficiency programs, and water recycling technologies to mitigate their footprint.
•**Location Considerations:** Semiconductor companies often choose locations for their fabs based on access to reliable power and abundant water supplies. Regions with stable infrastructure for electricity and water are preferred for setting up wafer fabs.

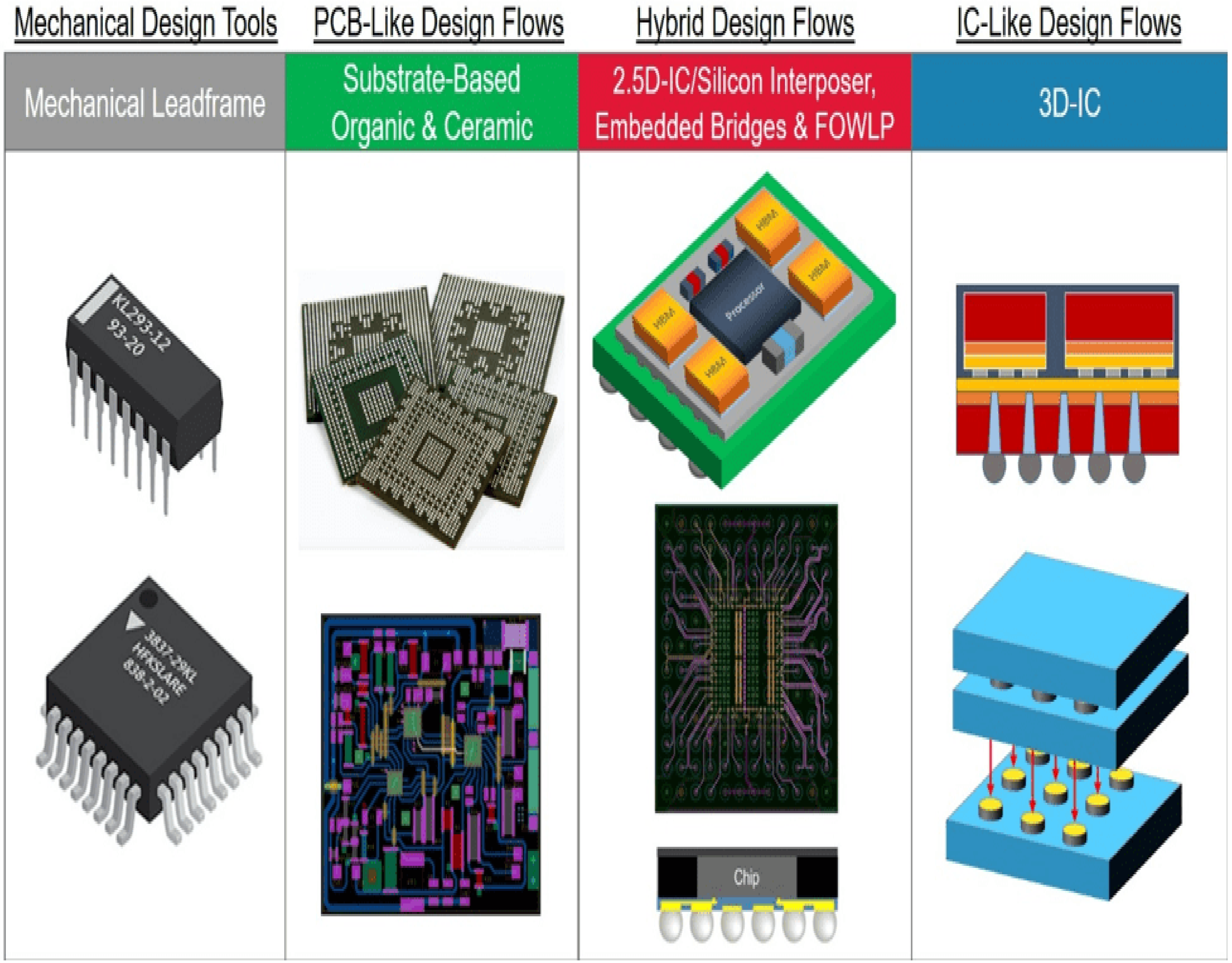
ATP

Assembly, Test, and Packaging



After front-end fabrication of the chips, wafers are typically sent to other facilities for back-end manufacturing activities such as assembly, test, and packaging (**collectively known as ATP**). In these steps, chips are cut from the silicon wafer, tested for performance, and packaged to protect the chip and to allow for its integration into finished electronic devices by attaching electrical interconnections. Multiple chips with different functions, such as microprocessors, graphics processors, and memory, are individually packaged and mechanically assembled on a printed circuit board, connected by wires or pins. Contract ATP manufacturers, are often referred to as Outsourced Semiconductor Assembly and Test (OSAT)

Advanced Packaging Techniques



Advanced packaging includes many different innovative techniques to improve chip-to-chip communications and satisfy increasing demand for integrating more diverse functionalities into smaller device footprints. Over time, as transistor density has increased, traditional interconnects between chips have limited the speed at which memory, logic, and other chips can communicate.

Many advanced packaging strategies have emerged to enable faster communications, including placing chips close to each other with a “bridge” (also referred to as a 2.5-dimensional, or 2.5D approach) or stacking them on top of one another (referred to as a three-dimensional [3D] strategy) to achieve the shortest distance between interconnections while decreasing the spatial footprint in the end device

EQUIPMENT

Capital Equipment

Semiconductor production requires complex equipment

- Photolithography machines to print circuit patterns;

- Etching and deposition equipment: Used for removing material or adding layers during the fabrication process.

- Testing machines: Critical for quality control and ensuring each chip meets specifications.

- U.S.-headquartered firms lead in the producing of nearly all types of semiconductor manufacturing equipment except **photolithography and wafer handling.**
- Chinese and Japanese companies lead in equipment used for wafer handling as well as packaging, assembly, and testing of semiconductors
- **Photolithography equipment is produced primarily in Europe (Netherlands) and Japan.**

Capital Equipment-Photolithography

Photolithography Equipment

Photolithography machines are an essential component of the semiconductor fabrication process. Photolithography equipment is used to pattern billions of electronic features on a silicon wafer by using different types of light passing through a mask (similar to a stencil) onto the underlying wafer, which is covered in light-sensitive materials. As electronic features have become smaller with more advanced process nodes, smaller wavelengths of light are required. For the most advanced process nodes (e.g., 3 nm and 5 nm), extreme ultraviolet (EUV) light is primarily used to pattern circuits onto the wafer; manufacturing processes for more mature process nodes (typically above 7 nm) use deep ultraviolet (DUV) light in the patterning process. EUV photolithography equipment has taken decades of research and development to produce and is made by a single company in the Netherlands, ASML. The remaining photolithography equipment market is accounted for by ASML and firms headquartered in Japan.

- https://www.youtube.com/watch?v=p4LqrdRahAo

Identify Companies Business Model Based on Activities

Integrated Device Manufacturer

Design, Fabrication, Testing

OSAT

Assembly, Testing

Foundry

Fabrication

Fabless IC Design

Design

EDA tool

Develop software to support various stages of IC design, including design entry, synthesis, simulation, verification, and physical implementation

Equipment Manufacturer

Design Equipment

Integrated Device Manufacturer

Foundry

EDA tool

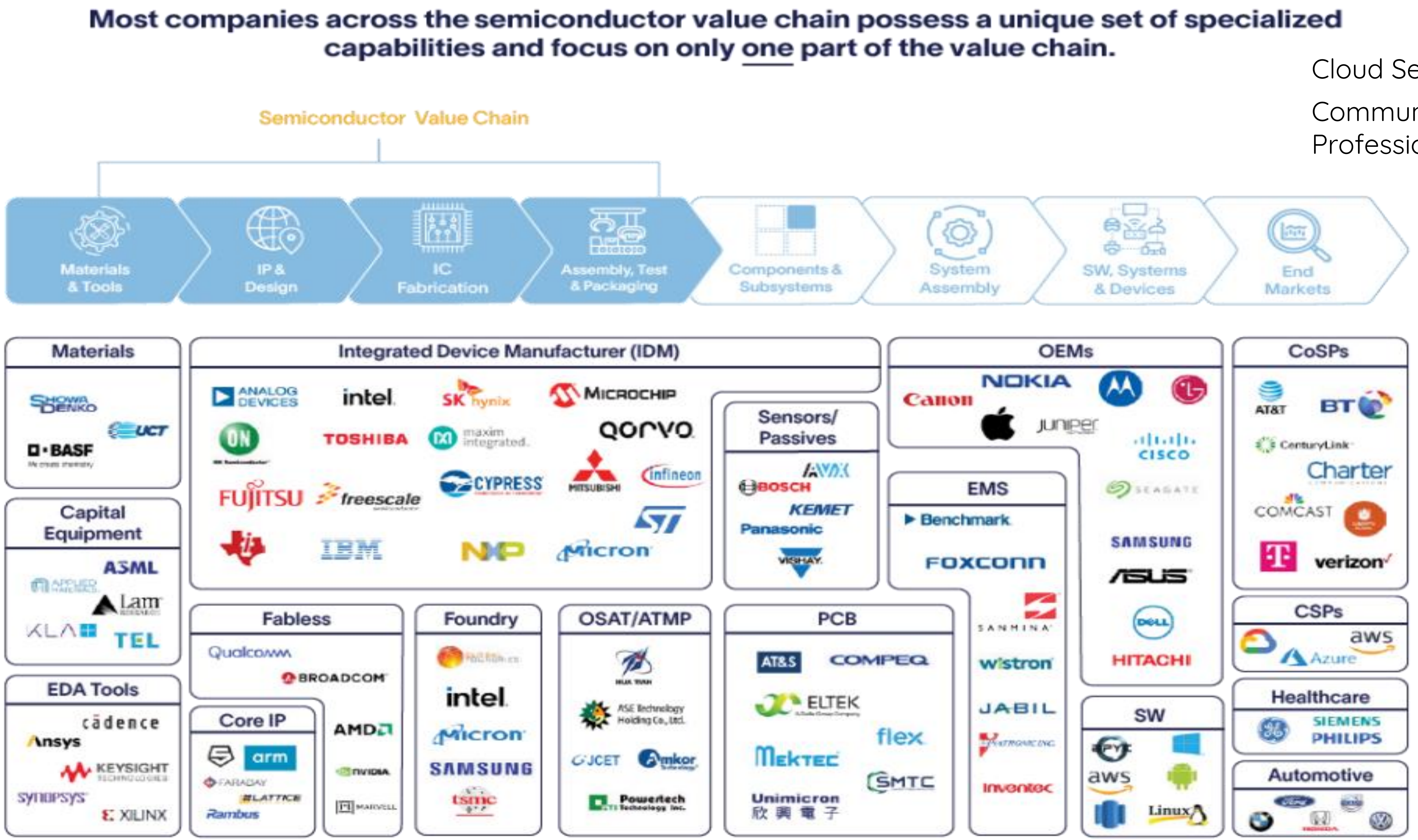
OSAT

Fabless IC Design

Equipment Manufacturer

SEMICONDUCTOR VALUE CHAIN

The end-to-end technology value chain requires tight integration between semiconductor companies, device manufacturers as well as system operators and end users



Cloud Service Provider (CSP)

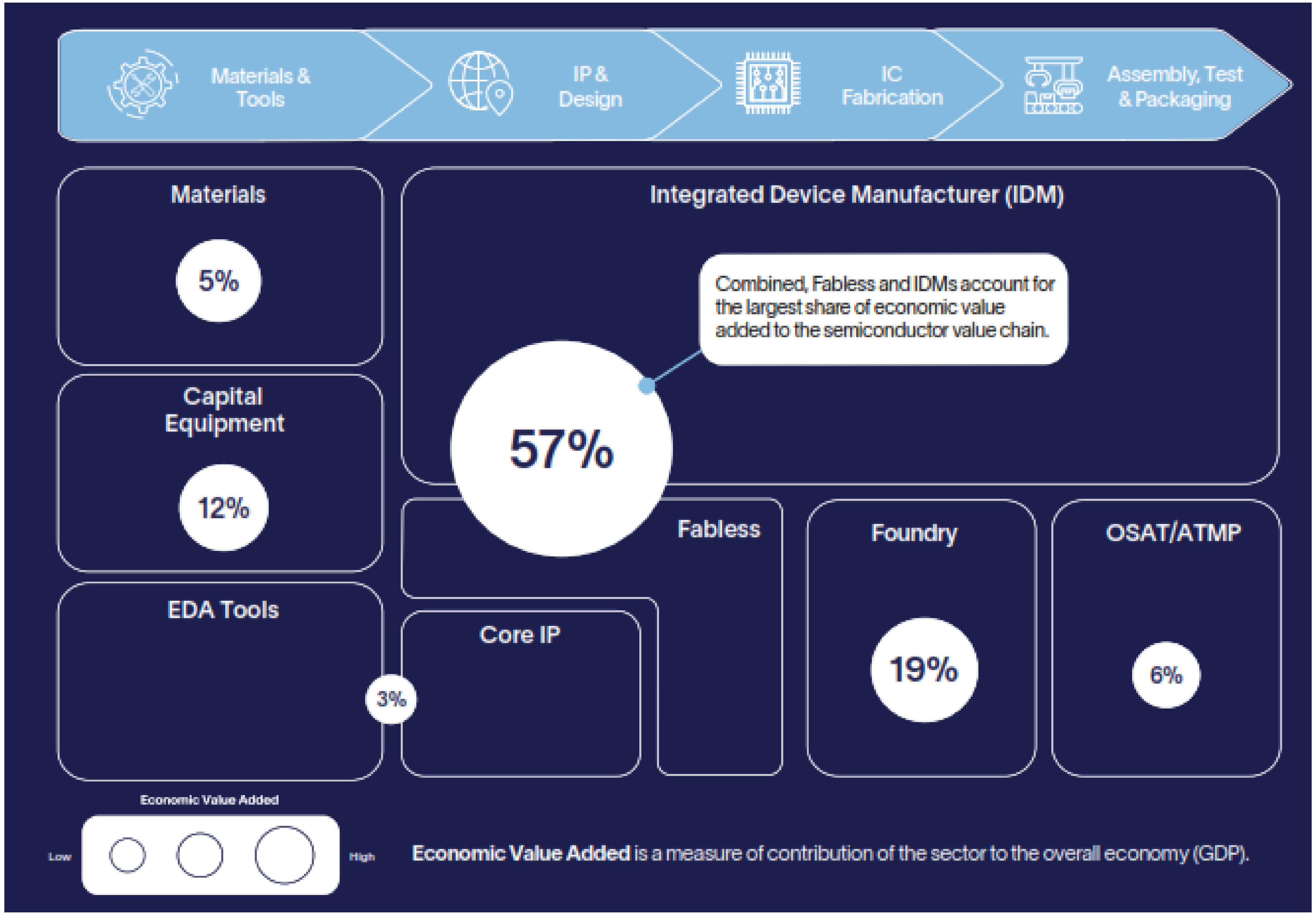
Communications Outsourcing and Professional Services=CoSP

- Fab-lite; company that own low tech wafer fab.
- Globalf oundry Spin by Motorola in 1999, acquired Fairchild in 2016
- Broadcom Internal Foundry acquired from Avago
- Intel Spin of Foundry as independent unit in sept 2024.

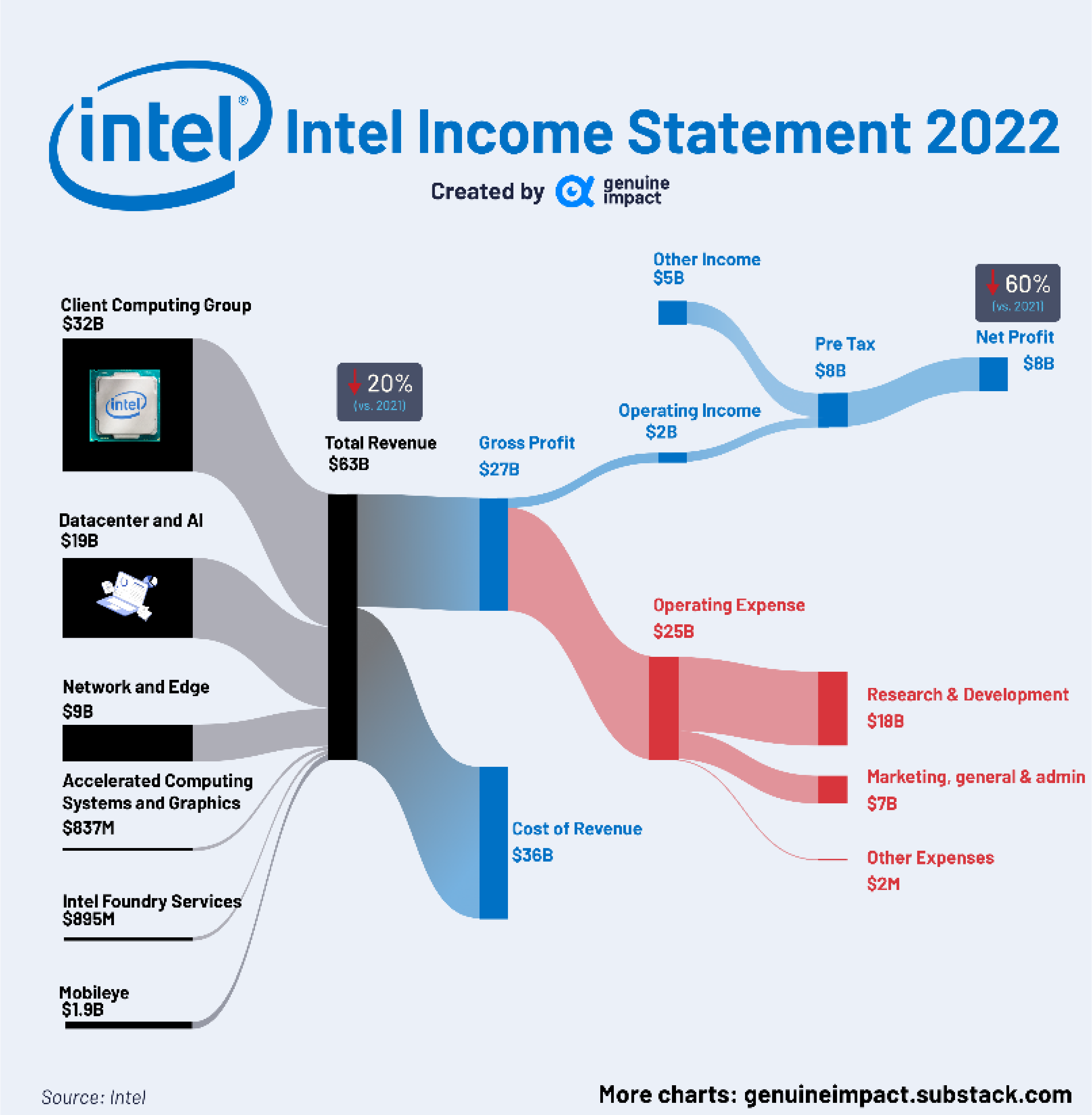
Ref: Altman Solon. (2022, November). The Semiconductor Ecosystem: Complex, Global, and Specialized.

The semiconductor value chain

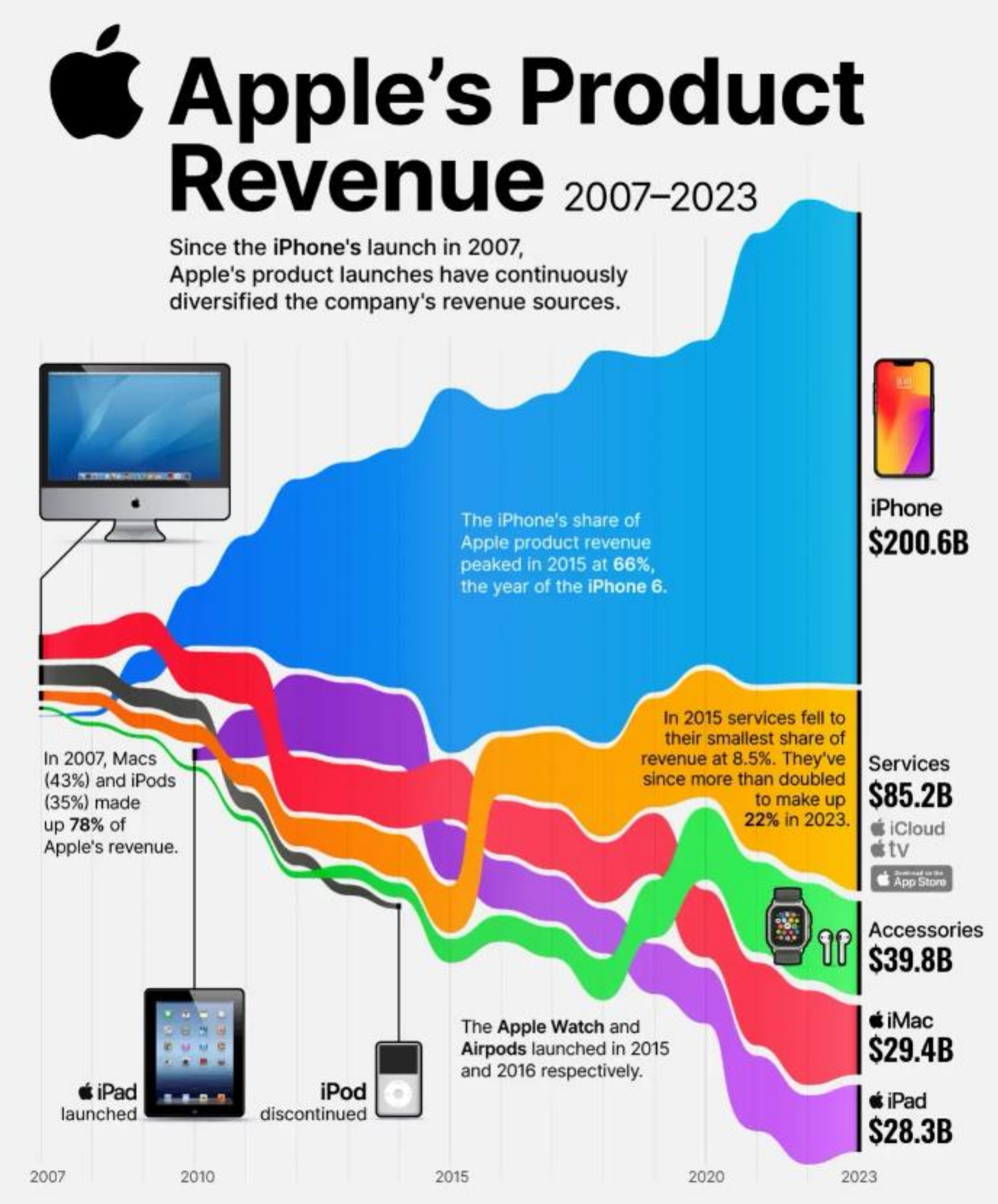
Fabless and IDMs, combined, account for the largest share of the economic value added in The semiconductor value chain



Revenue



Intel Income Statement 2022
Created by genuine impact
Client Computing Group
$32B
Datacenter and AI
$19B
Network and Edge
$9B
Accelerated Computing Systems and Graphics
$837M
Intel Foundry Services
$895M
Mobileye
$1.9B
20% (vs. 2021)
Total Revenue
$63B
Gross Profit
$27B
Cost of Revenue
$36B
Other Income
$5B
Operating Income
$2B
Pre Tax
$8B
60% (vs. 2021)
Net Profit
$8B
Operating Expense
$25B
Research & Development
$18B
Marketing, general & admin
$7B
Other Expenses
$2M
Source: Intel
More charts: genuineimpact.substack.com



Apple's Product Revenue 2007–2023
Since the iPhone's launch in 2007, Apple's product launches have continuously diversified the company's revenue sources.
The iPhone's share of Apple product revenue peaked in 2015 at 66%, the year of the iPhone 6.
iPhone
$200.6B
In 2015 services fell to their smallest share of revenue at 8.5%. They've since more than doubled to make up 22% in 2023.
Services
$85.2B
iCloud
tv
Accessories
$39.8B
iMac
$29.4B
iPad
$28.3B
In 2007, Macs (43%) and iPods (35%) made up 78% of Apple's revenue.
iPad launched
iPod discontinued
The Apple Watch and Airpods launched in 2015 and 2016 respectively.
2007
2010
2015
2020
2023

Another Design Landscape

As higher end applications demand more complex designs, a number of companies are moving away from in house IC design. Yet at the same time, the desire for higher performing chips have led other companies (e.g. Apple and Amazon) to ***design their own chips*** which are used in the iPhone, Mac computers and Amazon’s data centres. These cross currents create both opportunities and challenges for Taiwan’s leading IC design companies.

Engineering Disciplines

IDM	FOUNDRY	FABLESS	EDA	ATP*
Electronic Engineering (PhD level)	Semiconductor Physics (PhD level)	Electronic Engineering (PhD level)	Software Engineering (MSc, PhD level)	Electronic Engineering (MSc level)
Mechanical or Manufacturing Engineering (PhD level)	Electronic Engineering (PhD level)	Software Engineering (MSc level)	Electronic Engineering (MSc level)	Mechanical or Manufacturing Engineering (MSc level)
Semiconductor Physics (PhD level)	Mechanical or Manufacturing Engineering (PhD level)			Mechatronic Engineering (MSc level)
Chemical Engineering (PhD level)	Chemical Engineering(PhD level)			Software Engineering (MSc level)
Software Engineering (MSc level)	Software Engineering (MSc level)			

Outsourcing

PC Market			Mobile Market		
	Activities	In-house/Outsourced		Activities	In-house/Outsourced
Intel	Specification	Internal	Nokia	Specification	Internal
	Design	Internal		Design	External
	Verification	Internal and External		Verification	External
	Physical design	Internal and External		Physical design	External
	Manufacturing	Internal and External		Manufacturing	External
TI	Specification	Internal	Motorola	Specification	Internal
	Design	Internal and External		Design	Internal
	Verification	Internal and External		Verification	Internal and External
	Physical design	Internal and External		Physical design	Internal and External
	Manufacturing	Internal and External		Manufacturing	External
AMD	Specification	Internal	RIM	Specification	Internal
	Design	Internal		Design	Internal
	Verification	External		Verification	Internal
	Physical design	Internal		Physical design	Internal
	Manufacturing	External		Manufacturing	External
Samsung	Specification	Internal	Apple	Specification	Internal
	Design	Internal		Design	Internal
	Verification	Internal		Verification	Internal
	Physical design	Internal		Physical design	Internal
	Manufacturing	Internal		Manufacturing	External

The product development activities that can be outsourced can be broadly classified into specification, design, verification, physical design and manufacturing. *Many companies in the PC industry design their Integrated Circuits in-house to avoid technology leakage* and outsource verification and physical design work that consumes a greater portion of the project schedule - Nearly 70% of the product development cycle. *By outsourcing verification, the customers can better track verification costs and quality control over the skill* and knowledge of engineers devoted to verify complex systems and designs.

Cost & Expertise's, new product idea.

Ref : OUTSOURCING TRENDS IN SEMICONDUCTOR INDUSTRY
By
KARTHIKEYAN MALLI MOHAN

CHIPS Act

- Congress enacted the Creating Helpful Incentives to Produce Semiconductors (CHIPS) for America program through the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021 (2021 NDAA, P.L. 116-283).
- It authorized provisions for semiconductor manufacturing, research and development (R&D), workforce training and education, and collaboration and coordination with allied and other friendly countries.
- the CHIPS Act of 2022 (Division A of P.L. 117-167), signed into law by President Joe Biden on August 9, 2022, appropriated $52.7 billion to carry out these provisions for FY2022-FY2027.
- It includes $39 billion in financial incentives to expand domestic manufacturing capacities across the semiconductor supply chain, including for fabrication and advanced packaging, as well as suppliers of semiconductor equipment and materials.
- The act also appropriated $11 billion for R&D programs to promote U.S. leadership in producing next-generation semiconductor technologies.

Taiwan

Ecosystem in 2012

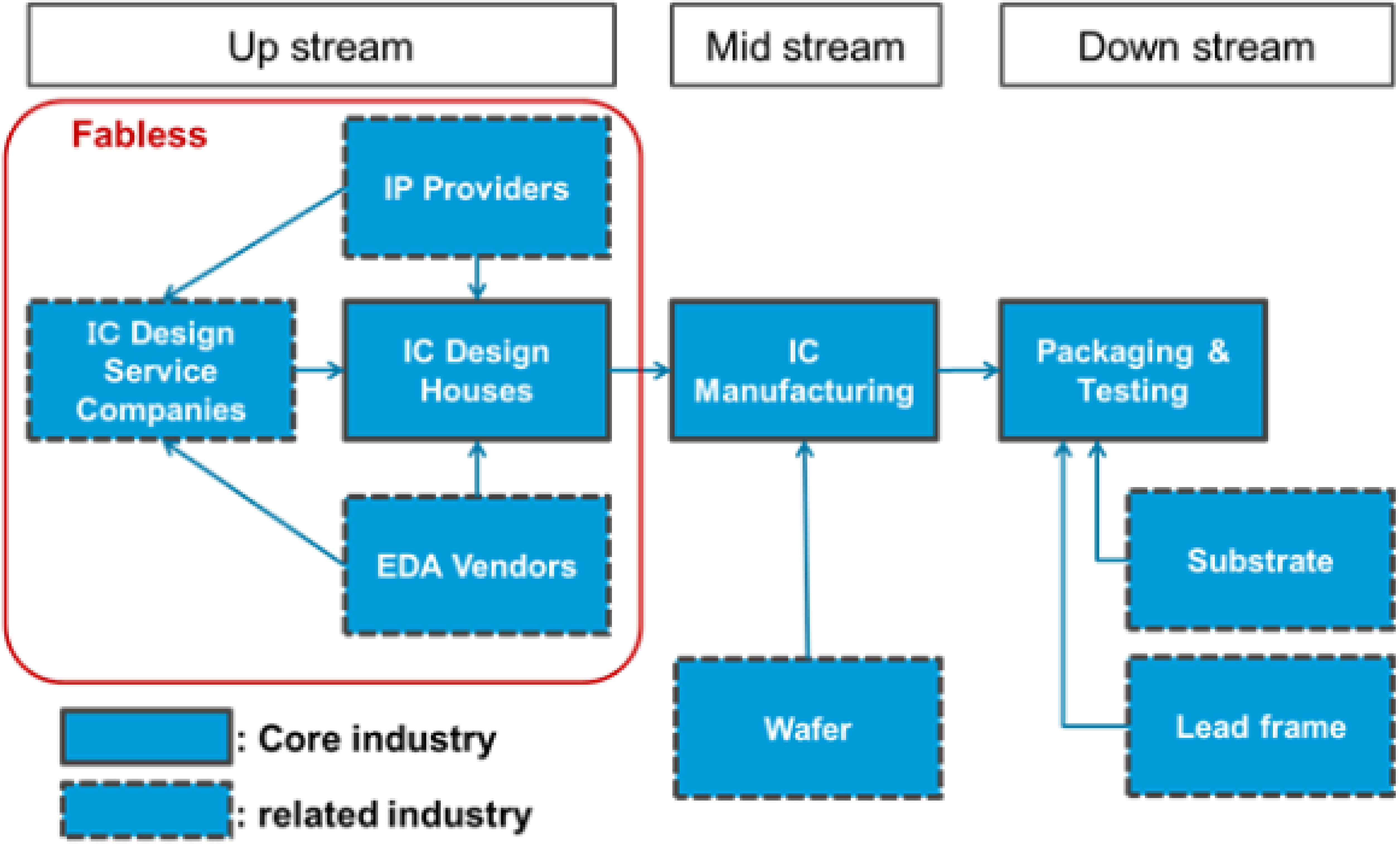
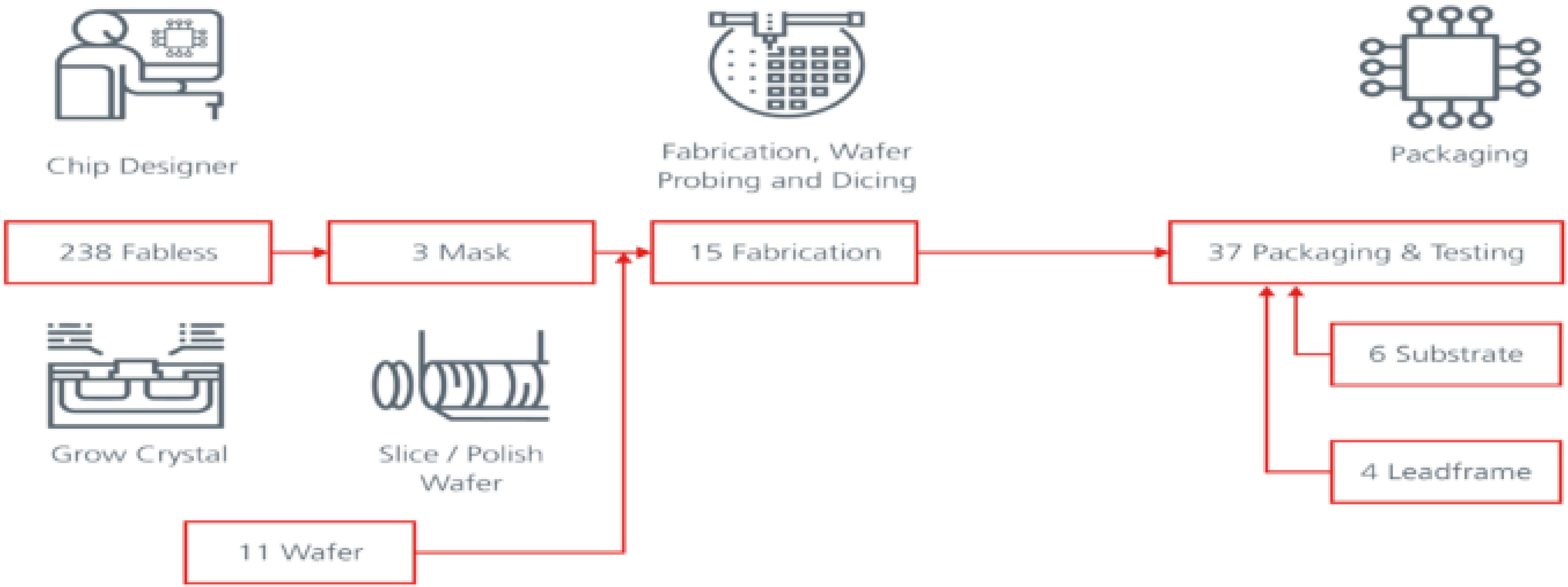


Fig. 1: Disintegrated infrastructure in Taiwan (2012)

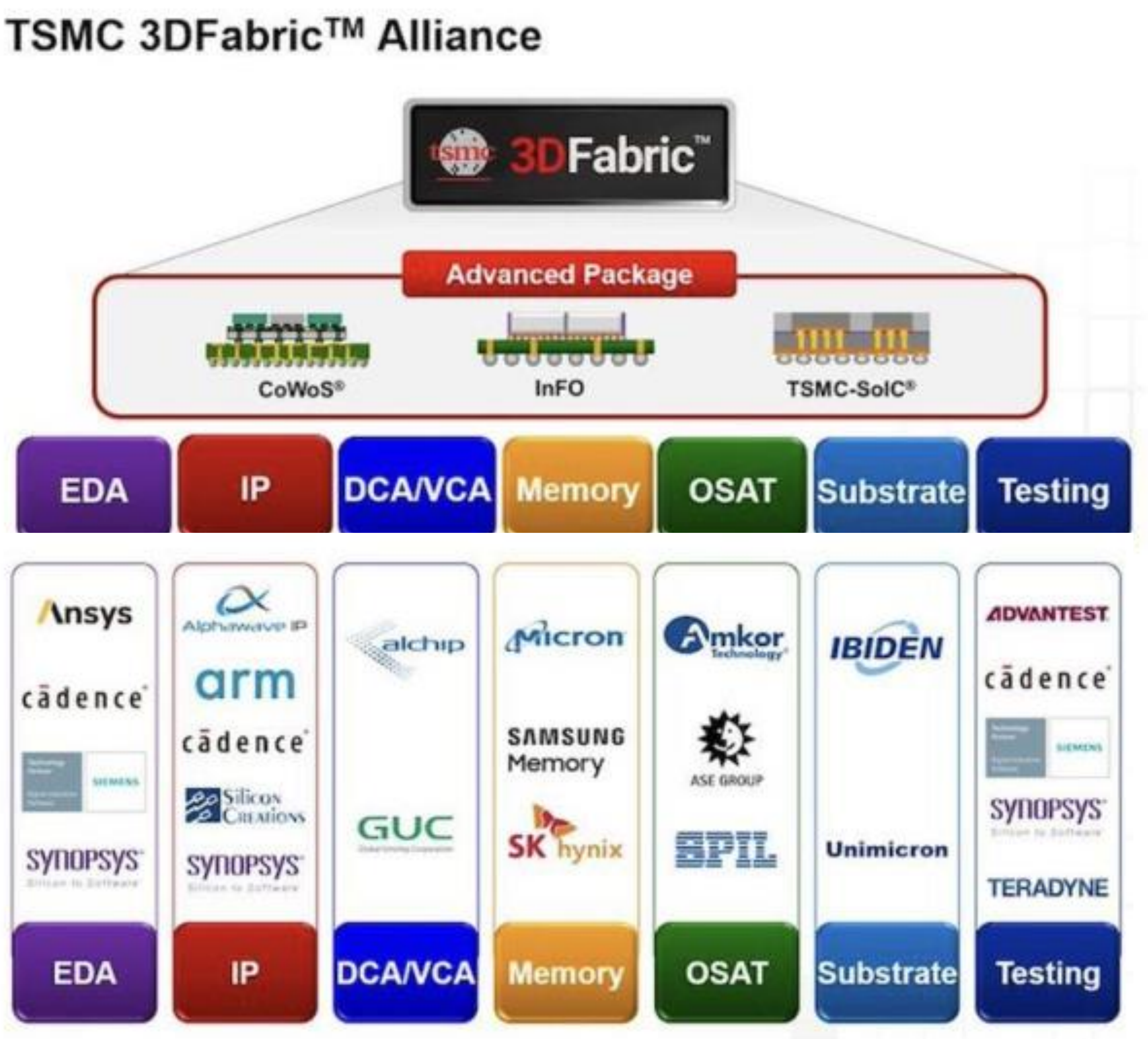
Ecosystem in 2020

Fig. 1. Taiwan IC ecosystem



Source: Overview on Taiwan Semiconductor Industry (2020 Edition). Taiwan Semiconductor Industry Association.

TSMC Advanced Packaging Alliance



Foundry Comparison

Table 1: Technology Trajectory of Lead National Firms

	Korea	Taiwan	Malaysia
	Samsung	TSMC	Silterra
1975	Started		
1976-1982	Acquisitions, Hirings and Licensing		
1983	64K DRAM		
1984	256K DRAM		
1986	1M DRAM, 1M SDRAM		
1987		Incorporated	
1988	4M DRAM	Hiring personnel and contract fabrication	
1990	16M DRAM		
1992	256M DRAM		
1996	1G DRAM		
1997			
1999	256MB NAND		
2000	516MB NAND		Started with 0.25CMOS
2001	1G NAND	Range of DRAMs, SDRAMs and SRAMs	0.22 and 0.18 CMOS
2002	2G NAND		
2003	4G NAND		
2004	8G NAND	12" wafer	
2005	16G NAND	NAND	8MB SRAM
2006	32G NAND		0.13 micron CMOS
2007	64G NAND		
2008			
2009	2 G DRAM	Microprocessors	

Note: Samsung also produces SDRAMs and SRAMs.

Source: Compiled from Kim (1997: 88), Rasiah, Kong, Lin and Song (2009); Authors' interviews (2007)

- 1st Silicon Opens Sarawak Wafer Fab-in 2000
- Infineon
- Global Foundries (Remote Manufacturing or Virtual Fab)

Semiconductor Challenges

History

Figure 1.0: ENIAC and the earliest transistor made in Tubes

Electronic Numerical Integrator and Computer (ENIAC), was the result of a U.S. government-funded project during World War II to build an electronic computer that could be programmed. The project was based out of the University of Pennsylvania's Moore School of Engineering.

Timeline

- 1940 Russel Ohl (PN junction)
  - The PN junction is developed at Bell Labs. The device produces 0.5 V across the junction when exposed to light.
- 1947 Bardeen and Brattain (Transistor)
  - 1945 Bell labs establish a group to develop an alternative to the vacuum tube. The group was lead by William Shockley.
  - Bardeen and Brattain succeeded in creating an amplifying circuit utilizing a point-contact "transfer resistance" device (the transistor).
  - The transistor was built on germanium.
- 1950 William Shockley (Junction transistor)
  - Higher and better manufacturability then the point-contact transistor is developed.
  - By the mid fifties the junction transistor replaces the point-contact transistor
  - Main use: telephone systems
- 1952 Single crystal silicon is fabricated
- 1954 First commercial silicon transistor
- 1954 First transistor radio (Regency TR-1)
  - Industrial Development Engineer Associates
  - Four germanium transistors from Texas Instruments
- 1955 First field effect transistor
  - Bell Labs

- Commercialization In mind, Led by PhD Holders within companies LABS.
- Motivation: to replace the vacuum tube
- Business model: licenses.
- William Shockley later was awarded the Nobel Prize in Physics in 1956 for his work on transistor. *The Nobel Prize in Physics 1956* was awarded jointly to William Bradford Shockley, John Bardeen and Walter Houser Brattain "for their researches on semiconductors and their discovery of the transistor effect“- eg. PN Junction use quantum mechanic.

Timeline

- 1952 Geoffrey W. A. Dummer (IC concept)
  - 1952 IC concept published
  - 1956 Failed attempt
- 1954 Oxide masking process developed
  - Developed at Bell Labs this is the foundation of IC production
  - The process involves: oxidation, photo-masking, etching and diffusion
- 1958 Jack Kilby (Integrated circuit)-Nobel Prize in 2000
  - Working at Texas Instruments Kilby built a simple oscillator IC with five integrated components
  - U. S. patent # 3,138,743 (1959)
- 1959 Planar technology invented
  - The planar technology was developed from the contributions of: Jean Hoerni and Robert Noyce (Fairchild) and Kurt Lehovec (Sprag Electric)
  - The planar technology is still the process used today.
- 1960 First MOSFET fabricated
  - At Bell Labs by Kahng
- 1961 First commercial ICs
  - Fairchild and Texas Instruments
- 1962 TTL invented
- 1963 First PMOS IC produced by RCA
- 1963 CMOS invented
  - Frank Wanlass at Fairchild Semiconductor
  - U. S. patent # 3,356,858
  - Standby power reduced by six orders of magnitude

The Importance of Academic

- Microelectronics research and instruction began early at UC Berkeley, soon after the invention of the integrated circuit in 1958. In 1960, Professors D. O. Pederson, T. E. Everhart, and P. L. Morton in Electrical Engineering conceived plans for an integrated circuits reserach laboratory, the first in the world at a university, which became reality in 1962.
- No Textbook IC until late 60s. By mid 70s an influential book was authored by Gray & Meyer
- Simulation Program with Integrated Circuit Emphasis (SPICE)
- Proper Analysis and Design
- Talents Pool

Master Level

- https://cockrell.utexas.edu/academics/graduate-education/programs/semiconductor-science-and-engineering

Track 1: Semiconductor Manufacturing

C - Advanced Physical Chemistry: Elements of Spectroscopy
C - Advanced Analytical Chemistry
C- Introduction to Computational Methods in Chemistry
C- Nanomaterials Chemistry and Engineering
E- Ultra-Large-Scale Integration Techniques
E- Synthesis, Growth and Analysis of Electronic Materials
E- Plasma Processing of Semiconductors I
E-396K.20 - Plasma Processing of Semiconductors II
ME -Practical Electron Microscopy
ME - Analytics and Control in Semiconductor Manufacturing
ME- Time-Series Modeling, Analysis and Control
ME - Optical Design
ME - Precision Machine Design

SSE-Semiconductor Science and Engineering

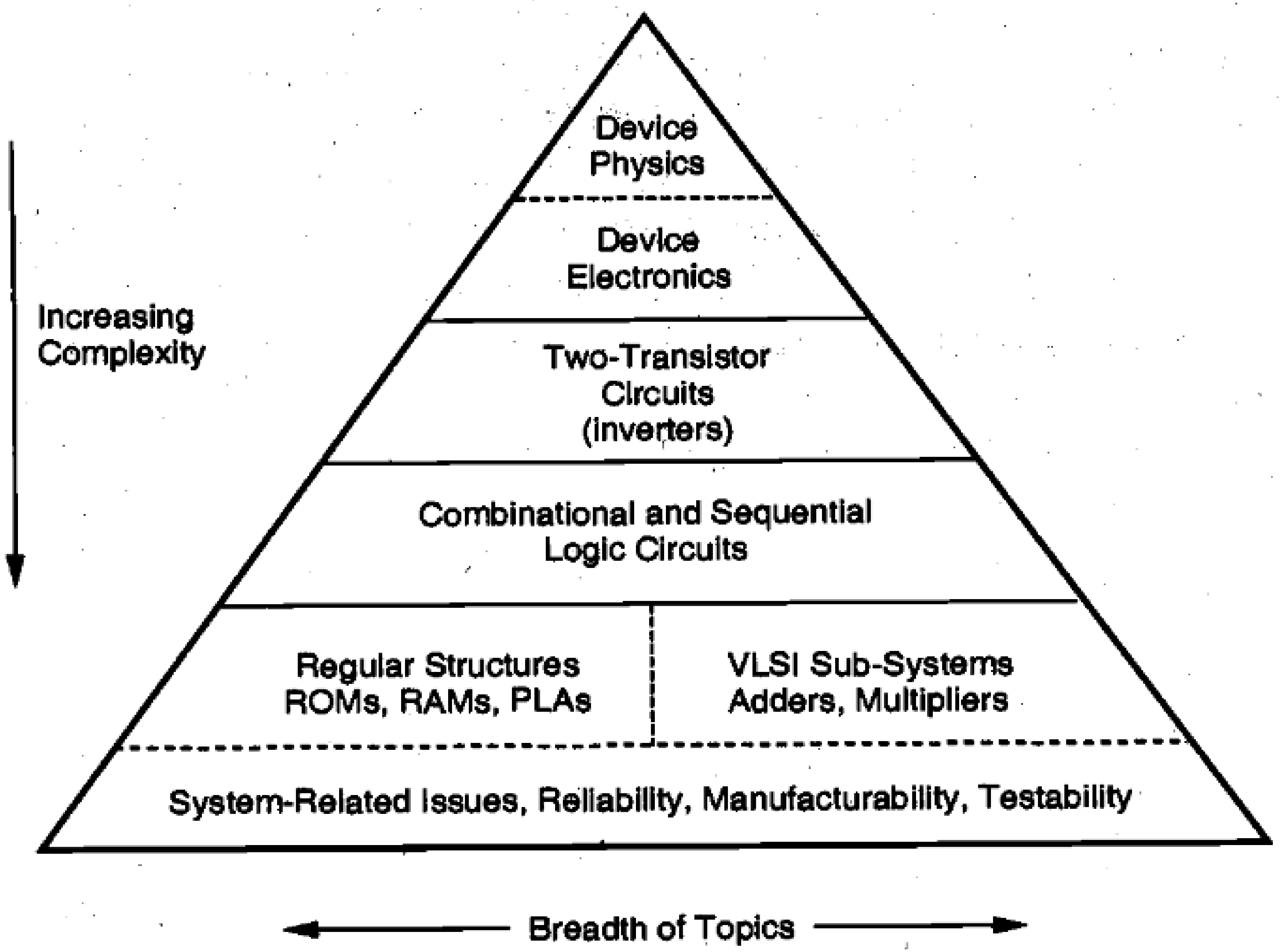
Track 2:: Heterogeneous Integration

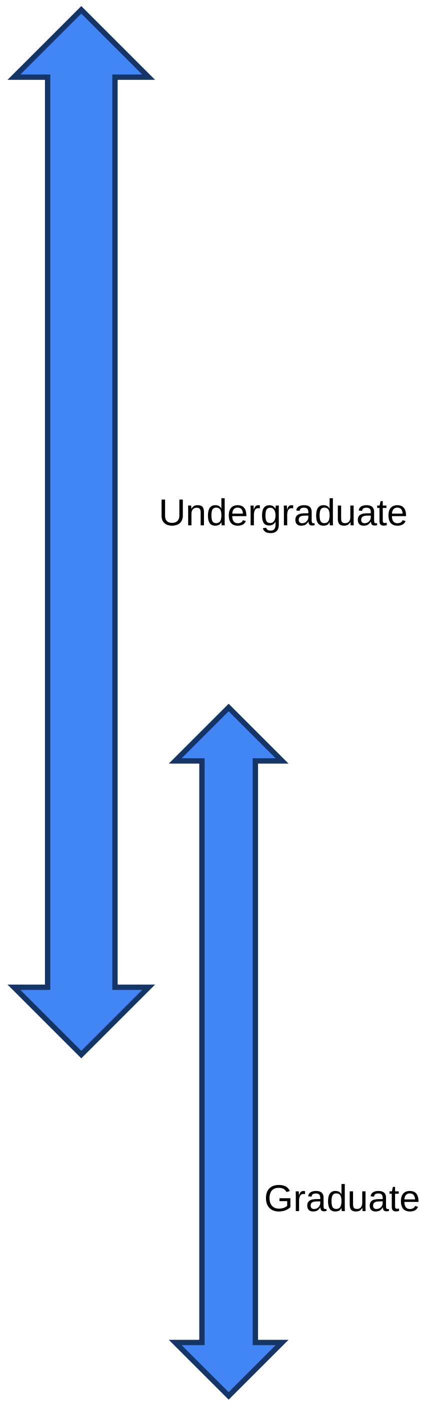
ECE - Optical Communication/ Interconnects
ME- Nanoscale Energy Transport and Conversion
SSE - Advanced Packaging and Heterogenous Integration
SSE- Microelectronics Packaging and Thermal Management
SSE- Materials for Semiconductor Packaging and Heterogeneous Integration
SSE- Failure and Reliability in Heterogeneous Integration

Track 3: Semiconductor Devices

ECE - Semiconductor Optoelectronic Devices
ECE - Semiconductor Heterostructures
ECE - Advanced Semiconductor Nanotechnology
ECE - Thin Film Transistors
ECE - Carbon and 2D Devices
ECE - Semiconductor Nanostructures
ECE - Microelectromechanical Systems
ECE - Quantum Theory of Electronic Materials
ECE - Power Semiconductor Devices
PHY - Density Functional Theory
PHY - Solid-State Physics I

Digital Integrated Circuit BOK





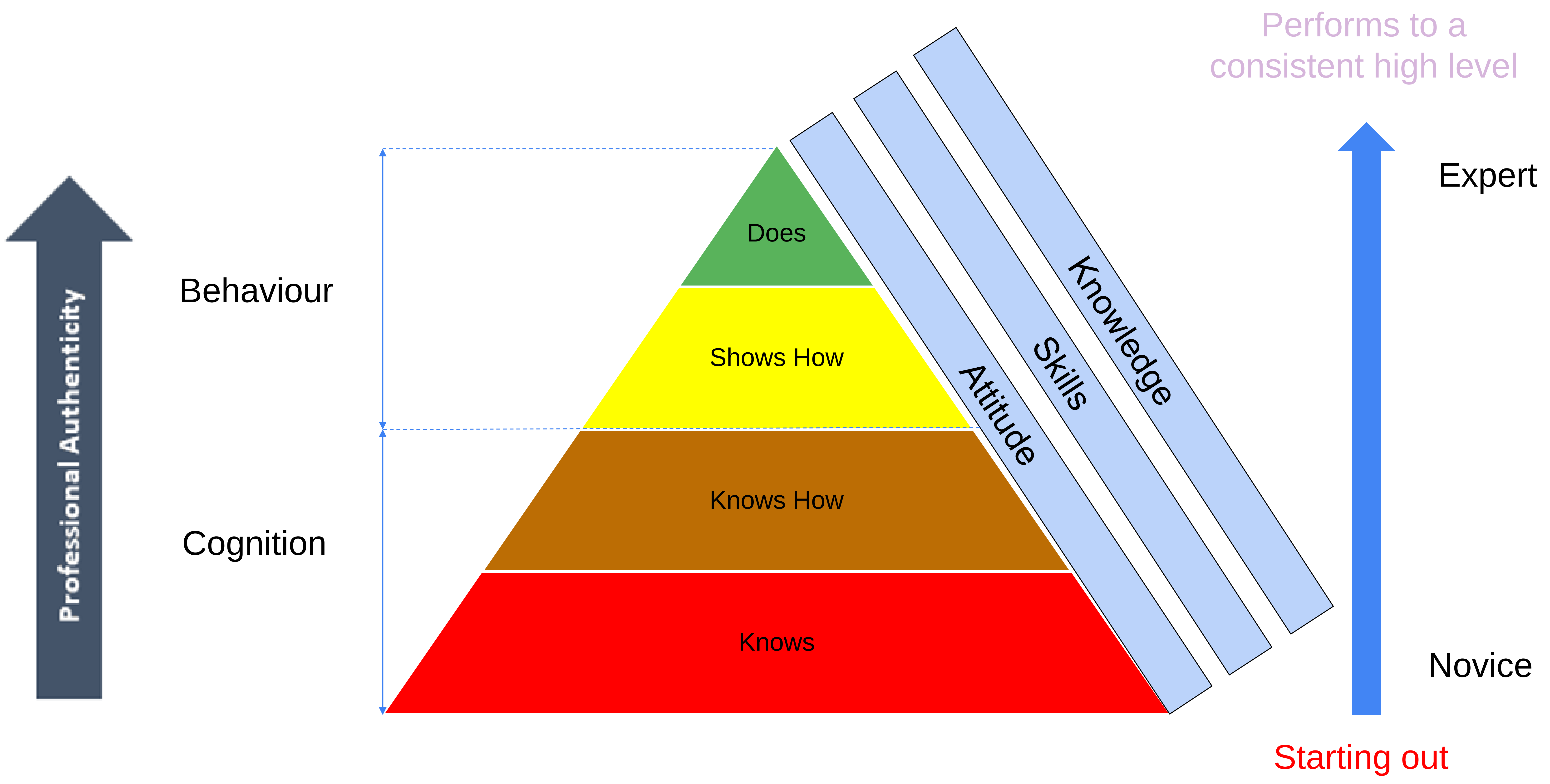
Analog Integrated Circuit BOK

1 Introduction to Analog Design
2 Basic MOS Device Physics
3 Single-Stage Amplifiers
4 Differential Amplifiers
5 Current Mirrors and Biasing Techniques
6 Frequency Response of Amplifiers
7 Noise
8 Feedback
9 Operational Amplifiers
10 Stability and Frequency Compensation
11 Nanometer Design Studies
12 Bandgap References
13 Introduction to Switched-Capacitor Circuits
14 Nonlinearity and Mismatch
15 Oscillators
16 Phase-Locked Loops
17. Short-Channel Effects and Device Models
18 CMOS Processing Technology
19 Layout and Packaging.

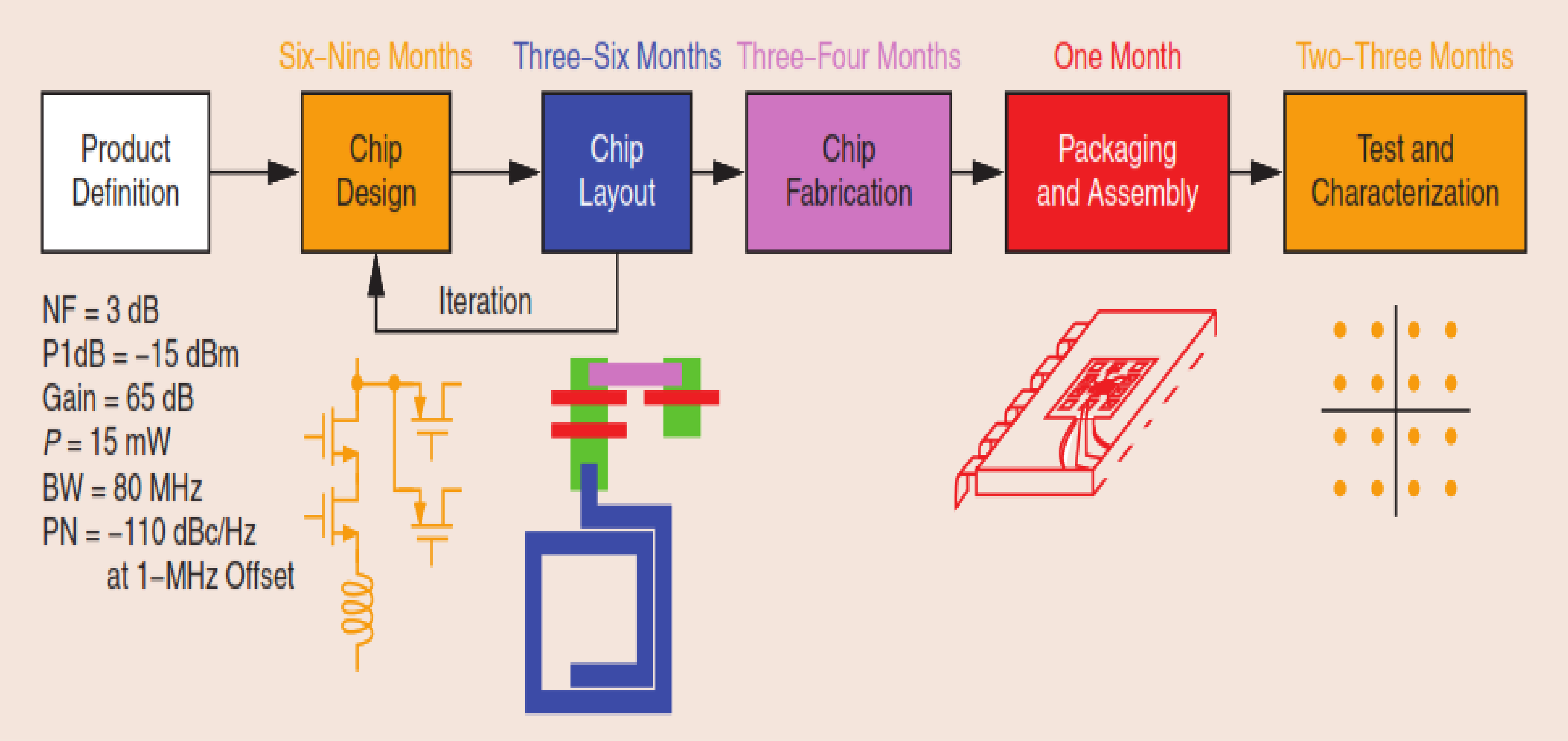
Undergraduate

Graduate

Developing and Assessing Competence



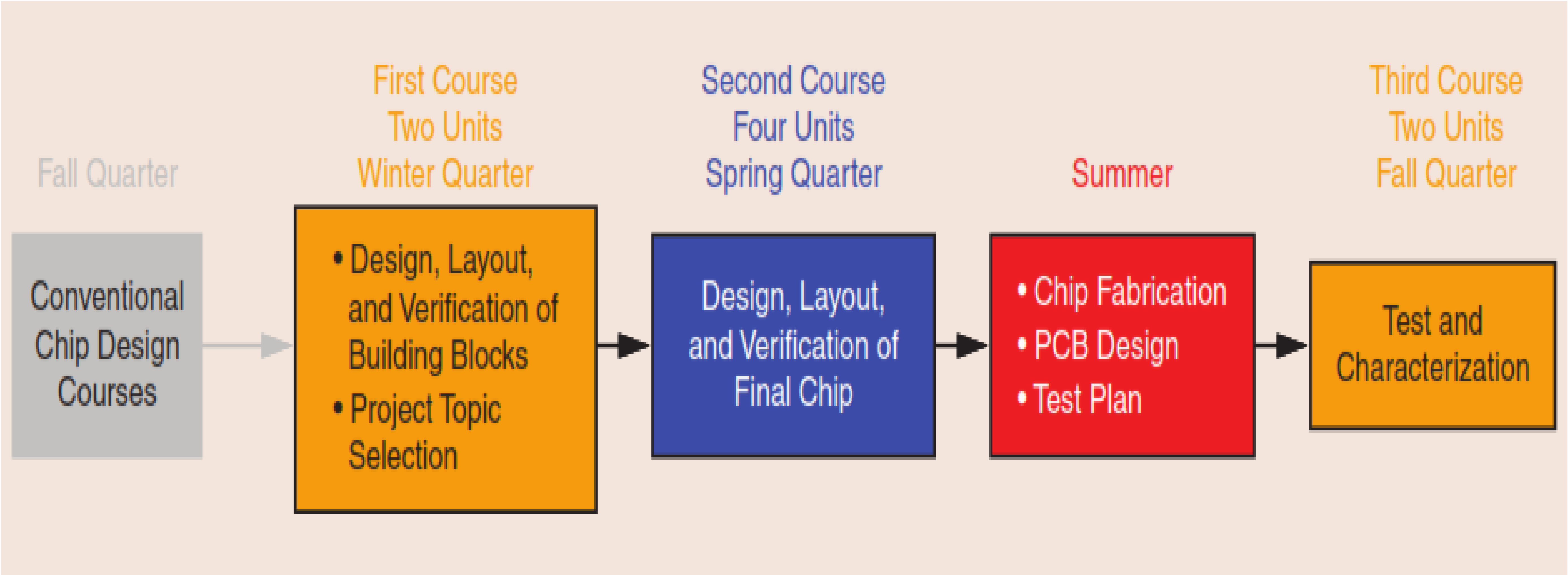
A typical chip development time frame.



One might argue that these missing components of chip design education can be learned after students enter the industry. However, product development cannot afford to place an inexperienced engineer in the critical path (Industry Attachment?). Thus, such training in the industry occurs only slowly and haphazardly.

Ref: Education of Chip Designers at a Large Scale: A Proposal

A structure of tapeout course



Ref: Education of Chip Designers at a Large Scale: A Proposal

The Industry needs ?

- Product meeting specific standards
- Application
- Technology

Summary

- The End-to-End Semiconductor Production Process Understanding
- Global Semiconductor Markets Comparison
- Semiconductor Industry Current and Future Direction

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Thank you!

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